AUTOMATED HELMET VIOLATION DETECTION AND VEHICLE NUMBER PLATE IDENTIFICATION USING DEEP LEARNING FOR INTELLIGENT TRAFFIC LAW ENFORCEMENT

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Dissertation submitted in partial fulfillment of the requirements for the Special Honours Degree of Bachelor of Science in Information Technology Specializing in Information Technology

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the Supervisor (Mr. Dinith Primal) Date

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ABSTRACT

Motorcycle helmet non-compliance remains a critical road safety concern, particularly in traffic environments where manual monitoring is inefficient, inconsistent, and difficult to scale. This research component presents an automated helmet violation detection and vehicle number plate identification subsystem developed for an intelligent traffic violation management framework. The proposed system uses a YOLOv8-based computer vision model to detect helmet compliance from traffic surveillance footage and classify riders without helmets as violation cases. When a violation is detected, the system captures visual evidence, localizes the associated vehicle number plate using a YOLOv8-based plate detection model, and extracts the plate information through optical character recognition. In compliant cases, number plate recognition is not performed, ensuring that vehicle identification is triggered only for confirmed violations.

The experimental results demonstrate that the subsystem can identify helmet violations, generate evidence, and associate detected violations with vehicle registration information. The number plate detection module achieved strong localization performance, while no-helmet detection remained more challenging due to factors such as occlusion, motion blur, camera angle, and small head-region features. Overall, the proposed subsystem provides a practical foundation for automated, evidence-based traffic law enforcement within intelligent transportation systems.

Keywords – Helmet violation detection, number plate recognition, YOLOv8, deep learning, computer vision, traffic surveillance, automated traffic enforcement, optical character recognition.

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I wish to acknowledge the assistance received from my fellow research group members for their cooperation, discussions, and support during the development of the integrated traffic violation management framework. Their contributions to the broader system components helped establish the collaborative foundation within which my individual component, the helmet violation detection and vehicle number plate identification subsystem, was developed.I further acknowledge the support received in the collection and preparation of traffic-related images, videos, and other materials required for model training, testing, and evaluation. The assistance provided in reviewing system outputs, identifying technical issues, and validating the functional behavior of the proposed component contributed significantly to the improvement of the research outcome.

The design, implementation, testing, data analysis, and preparation of this dissertation were carried out by me, with academic supervision and technical guidance provided where necessary. The development of the helmet violation detection model, vehicle number plate identification pipeline, evidence generation process, and related documentation represents my individual contribution to the overall research project.

This research was not externally funded. Finally, I would like to thank my family and friends for their constant encouragement, patience, and support throughout the research period. Their motivation and understanding greatly assisted me in completing this dissertation successfully.

Table of Contents

LIST OF FIGURES

LIST OF TABLES

CODE LISTING

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
CCTV	Closed-Circuit Television
CNN	Convolutional Neural Network
CV	Computer Vision
DFL	Distribution Focal Loss
DL	Deep Learning
FPS	Frames Per Second
GPU	Graphics Processing Unit
GUI	Graphical User Interface
ITS	Intelligent Transportation System
YOLO	You only look once
mAP	Mean Average Precision
OCR	Optical Character Recognition
OpenCV	Open Source Computer Vision Library
XML	Extensible Markup Language

1. INTRODUCTION

Road traffic safety continues to be a major public health, transportation, and governance concern across the world. According to the World Health Organization, road traffic crashes account for approximately 1.19 million deaths annually, while many more individuals suffer non-fatal injuries that often result in long-term physical, social, and economic consequences [1]. Within this broader safety landscape, motorcyclists constitute one of the most vulnerable road user groups, largely because they lack the structural protection available to occupants of enclosed vehicles. As a result, rider safety depends heavily on compliance with protective measures, among which helmet use remains one of the most critical. The World Health Organization further notes that correct helmet use can reduce the risk of death in a crash by more than six times and the risk of brain injury by up to 74%, thereby establishing helmet enforcement as a central road safety priority rather than a minor regulatory concern [2].

Despite the recognized importance of helmet legislation, the practical enforcement of motorcycle safety regulations remains difficult in many real-world traffic environments. Traditional monitoring approaches are still heavily reliant on manual observation, roadside inspection, and human judgment, all of which are resource-intensive, difficult to scale, and often inconsistent under dense or continuous traffic conditions. In urban road networks characterized by high vehicle volume, limited enforcement personnel, and the need for rapid response, purely manual mechanisms are insufficient to ensure persistent monitoring of rider behavior. Consequently, there is an increasing need for automated and intelligent traffic surveillance solutions capable of identifying violations accurately, preserving evidence systematically, and supporting timely enforcement action.

Recent advances in artificial intelligence and computer vision have created significant opportunities to address these limitations. In particular, deep learning-based methods have transformed the way traffic scenes are analyzed by enabling automatic detection, classification, tracking, and interpretation of road users and traffic events. Contemporary research in intelligent traffic surveillance increasingly emphasizes the use of vision-based systems for tasks such as vehicle detection, anomaly analysis, event recognition, and traffic behavior monitoring [3], [4]. These developments have made it possible to shift from passive video recording toward active, data-driven traffic monitoring systems that can interpret visual information and generate actionable outputs. However, while substantial progress has been made in generic traffic perception, the practical deployment of such techniques for specific law-enforcement tasks still presents important research challenges, particularly under uncontrolled roadside conditions involving occlusion, motion blur, background clutter, and variable illumination [3], [4].

Within this context, motorcycle helmet violation detection has emerged as an important area of applied computer vision research. Prior studies have demonstrated that deep learning models can be trained to identify riders and determine helmet usage from traffic images and videos with promising levels of accuracy [5], [6]. This body of work confirms that helmet compliance analysis is technically feasible and can support broader road safety initiatives through faster and more scalable observation. Nevertheless, the academic and operational significance of helmet violation detection extends beyond simply determining whether a rider is wearing a helmet. For such a system to be genuinely useful in a law-enforcement environment, the detected violation must be transformed into verifiable and traceable evidence that can support identification of the offending vehicle and subsequent regulatory action.

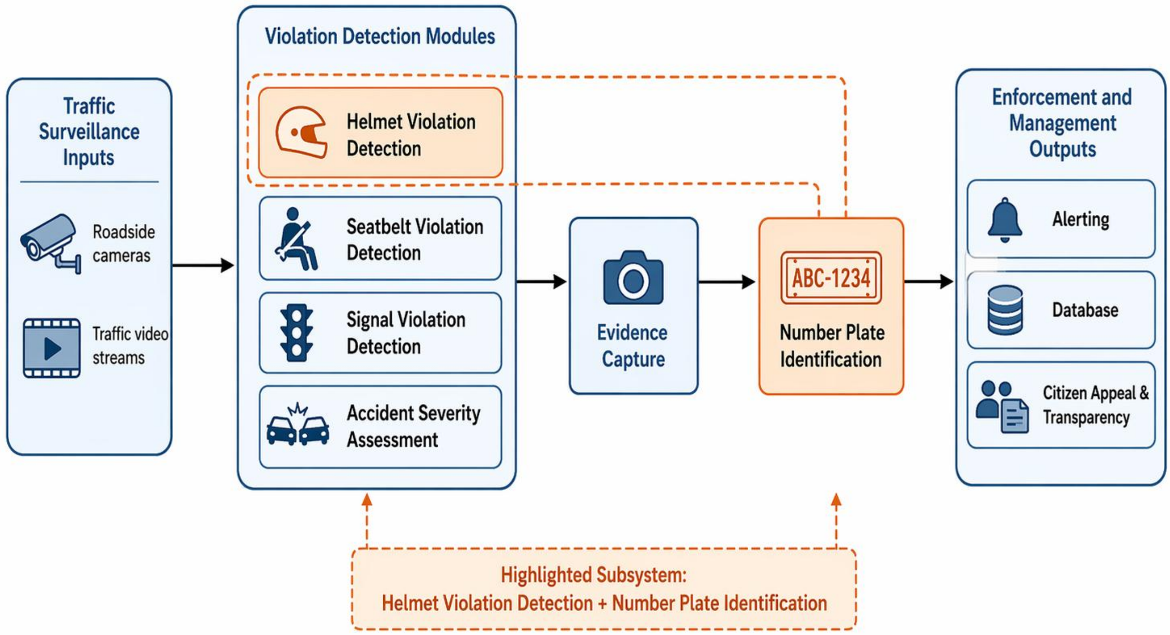
This requirement introduces the second essential dimension of the present research: vehicle number plate identification. Number plate detection and recognition are fundamental enabling technologies in intelligent transportation systems because they allow a visual event captured in traffic footage to be associated with a specific vehicle identity. Existing research on Automatic Number Plate Recognition (ANPR) has produced a wide range of methods for plate localization, character segmentation, and text recognition, and deep learning has significantly improved performance in this domain [7], [8]. Even so, number plate identification in surveillance imagery remains challenging because plates are often captured under non-ideal conditions, including low resolution, angular distortion, motion-induced blur, inconsistent lighting, and partial occlusion [7]. These limitations are especially important in enforcement-oriented applications, where identification reliability directly affects the practical value of the detection pipeline.

When the helmet detection literature and the number plate recognition literature are considered together, an important gap becomes visible. A considerable portion of existing work focuses either on detecting rider behavior or on recognizing vehicle identity, but fewer studies connect these capabilities within a unified enforcement-support process. In other words, many systems are able to identify a traffic-related visual condition, yet comparatively fewer are designed to convert that visual observation into a structured workflow involving violation confirmation, evidence capture, vehicle identification, and integration with a broader traffic management platform. This gap is particularly relevant in the case of motorcycle helmet violations, where effective enforcement requires not only the recognition of rider non-compliance but also the ability to associate that event with the corresponding vehicle.

The present research is situated precisely at this intersection. It forms part of a broader intelligent traffic violation management framework and focuses specifically on the design and implementation of a helmet violation detection and vehicle number plate identification subsystem. The subsystem is intended to process traffic surveillance footage in order to determine whether motorcycle riders comply with mandatory helmet-wearing regulations. When a rider is identified as not wearing a helmet, the system classifies the event as a violation, captures the relevant visual evidence, and proceeds to identify the associated vehicle through number plate detection and recognition. In this way, the subsystem contributes not merely to traffic scene understanding, but to a more practical and operationally meaningful objective: enabling automated, evidence-driven traffic law enforcement.

From an academic perspective, this subsystem is significant for several reasons. First, it addresses a socially important safety problem with direct implications for injury prevention and regulatory compliance. Second, it applies contemporary deep learning techniques to a real-world surveillance scenario in which environmental variability and operational constraints are unavoidable. Third, it moves beyond isolated detection by linking helmet non-compliance to vehicle identification, thereby increasing the actionable value of the visual analysis. Finally, because the subsystem is developed within an integrated traffic violation management architecture, it contributes to the broader research goal of building intelligent, scalable, and transparent digital enforcement systems capable of supporting modern transportation governance.

Figure 1.1. Conceptual Context Diagram of the Overall Traffic Violation Management Framework



1.1 Background and Literature Survey

1.1.1 Road safety, helmet compliance, and the need for automated enforcement

Helmet wearing is one of the most widely recognized and evidence-based interventions for reducing the severity of motorcycle-related injuries. However, the effectiveness of helmet laws depends not only on legislative presence but also on sustained enforcement. WHO guidance emphasizes that when traffic laws addressing helmets and other risk factors are not actively enforced, compliance declines and the expected safety benefit is significantly reduced [1], [2]. In practice, therefore, the challenge is not merely to define safety regulations, but to operationalize monitoring mechanisms capable of identifying violations consistently across time and location.

Conventional roadside enforcement techniques are often episodic, officer-dependent, and limited by resource constraints. In high-density road environments, these limitations become more severe because continuous manual monitoring is rarely feasible. This has motivated a shift toward surveillance-based and data-driven monitoring frameworks, especially in smart city and intelligent transportation contexts, where automated visual analysis can support persistent observation, rapid event detection, and digitally preserved evidence for downstream processing [3], [5].

1.1.2 Intelligent traffic surveillance and deep learning in traffic systems

Recent advances in artificial intelligence, particularly deep learning, have significantly reshaped the field of traffic video analysis. Contemporary traffic surveillance research increasingly relies on deep neural network architectures for object detection, classification, tracking, event interpretation, and anomaly analysis. Razi *et al.* describe a modern traffic video analysis pipeline as comprising stages such as video enhancement, segmentation, object detection, trajectory extraction, speed estimation, event analysis, and anomaly detection, illustrating the breadth of deep learning applications in traffic safety analysis [3]. Similarly, recent survey work on traffic surveillance systems shows that vision-based monitoring now encompasses both low-level tasks, such as detection and tracking, and high-level tasks, such as behavior understanding and event reasoning [5].

At the same time, the literature also shows that strong benchmark performance does not automatically translate into reliable field deployment. Azfar *et al.* note that real-world traffic environments introduce persistent challenges such as irregular lighting, occlusion, heterogeneous road behavior, diverse camera viewpoints, data bias, and computational constraints, all of which complicate practical deployment [4]. Zhou *et al.* further identify perceptual degradation, semantic understanding gaps, sensing coverage limitations, and computational resource demands as key unresolved issues in traffic surveillance systems [5]. These findings are highly relevant to violation monitoring tasks, which must function under precisely such uncontrolled real-world conditions.

1.1.3 Helmet violation detection for motorcyclists

Research on automated helmet detection has evolved from conventional image processing and handcrafted feature methods toward deep learning-based detection frameworks. Early large-scale work by Siebert and Lin demonstrated the feasibility of automated motorcycle helmet use analysis from roadside video, training a system on 91,000 annotated frames collected across multiple observation sites in seven cities in Myanmar and showing that helmet use rates could be estimated with high agreement relative to human observers [6]. This work was important because it established helmet-use detection as a meaningful traffic safety analytics task rather than a narrow object recognition exercise.

Subsequent research has focused on improving robustness under more challenging visual conditions. Chen *et al.* proposed a deep learning approach using residual transformer-spatial attention for motorcycle helmet detection and reported strong performance in aerial photography scenes, highlighting the value of enhanced feature extraction when riders appear at small scales or in visually complex backgrounds [7]. More recently, Shoman *et al.* employed YOLOv8 together with DCGAN-generated synthetic images to improve helmet violation detection, particularly under class imbalance and multi-rider conditions, and reported improved

F1-score relative to a stand-alone YOLOv8 model [8]. Other recent work similarly emphasizes that helmet detection in realistic traffic scenes remains vulnerable to missed detections and misdetections due to small targets, cluttered backgrounds, and occlusion, even when modern one-stage detectors are used [8], [13].

Taken together, the helmet-detection literature shows that automated helmet monitoring is technically feasible and socially meaningful. However, much of the work remains focused on detection accuracy itself, rather than on how detected non-compliance can be translated into a structured, evidence-supported enforcement action. In other words, identifying a rider without a helmet is only one part of the broader traffic governance problem.

1.1.4 Number plate detection and recognition in traffic surveillance

Automatic Number Plate Recognition (ANPR), also referred to in some literature as Automatic License Plate Recognition (ALPR), is a key enabling technology in intelligent transportation systems because it supports vehicle identification for law enforcement, parking management, tolling, and access control. A detailed survey by Lubna, Mufti, and Shah describes ANPR as a multi-stage problem involving image acquisition, plate localization, character segmentation or segmentation-free reading, and character recognition, while also highlighting the diversity of algorithms and implementation challenges across stages [9].

Recent deep learning research has improved both plate localization and text extraction. Ammar *et al.* proposed a multi-stage deep-learning-based vehicle and license plate recognition system with real-time edge inference, illustrating the growing interest in deployable end-to-end vehicle identification pipelines rather than isolated recognition modules [10]. Likewise, Elihos *et al.* addressed the problem of license plate recognition from roadway surveillance cameras and emphasized that high-quality plate recognition is indispensable for automated traffic enforcement, while also noting that general license plate recognition systems often require high-resolution imagery for strong performance [11]. More recent work by Vargoorani and Suen similarly shows that plate detection and recognition remain sensitive to variable lighting, regional font differences, and dataset-specific characteristics, even under deep learning approaches [12].

These findings are directly relevant to surveillance-based traffic enforcement. In practical traffic scenes, number plates may appear small, blurred, angled, partially occluded, or visually degraded by lighting and motion. Consequently, although deep learning has significantly advanced ANPR performance, reliable vehicle identification in uncontrolled traffic environments remains a non-trivial challenge. This is particularly important when plate reading is expected to support legal or administrative action following a detected violation.

1.1.5 Toward integrated enforcement-support pipelines

An important direction in recent research is the movement from isolated perception tasks toward integrated enforcement or monitoring workflows. Survey literature increasingly frames traffic surveillance as a layered process in which low-level visual perception should support higher-level interpretation, decision-making, and operational deployment [3], [5]. Similarly, multi-stage vehicle and plate-recognition systems demonstrate that practical traffic intelligence increasingly depends on connecting detection, recognition, and downstream processing in a coherent pipeline [10].

However, when the literature on helmet detection and the literature on ANPR are considered together, a clear pattern emerges: many studies address either rider compliance detection or number plate recognition, but comparatively fewer works operationalize these tasks as a unified enforcement-support workflow. In particular, the literature contains fewer studies that explicitly combine helmet non-compliance detection, evidence capture, and vehicle identification within one surveillance-driven subsystem capable of contributing to a broader traffic violation management framework. This unresolved intersection provides the intellectual foundation for the present study.

Table 1.1. Summary of Selected Prior Studies on Helmet Detection and Number Plate Recognition

Author / Year	Focus	Method	Strength	Limitation
WHO, 2023 [1], [2]	Road safety and protective compliance	Global road safety reporting and guidance	Establishes the public safety importance of helmet compliance	Does not address technical implementation of automated detection
Razi et al., 2022 [3]	Deep learning for traffic safety analysis	Review of deep learning-based traffic video analysis	Provides a broad framework for intelligent traffic surveillance tasks	General review, not specific to helmet-based enforcement
Azfar et al., 2022 [4]	Computer vision in complex traffic environments	Review of deep learning perception methods	Highlights real-world deployment challenges such as occlusion and lighting	Focuses on general perception rather than a specific enforcement workflow
Siebert and Lin, 2020 [5]	Helmet use detection	Deep learning on roadside video	Demonstrates feasibility of automatic helmet-use analysis at scale	Oriented more toward detection and analysis than enforcement linkage

Author / Year	Focus	Method	Strength	Limitation
Shoman et al., 2024 [6]	Helmet violation detection	YOLOv8 with synthetic data support	Improves violation detection under imbalance and multi-rider conditions	Primarily centered on detection performance
Lubna et al., 2021 [7]	Number plate recognition	Survey of ANPR methods	Comprehensive overview of localization and recognition approaches	Survey-oriented, not an integrated system study
Ammar et al., 2023 [8]	Vehicle and plate recognition	Multi-stage deep learning with real-time inference	Demonstrates practical end-to-end plate-related system design	Not specifically focused on helmet non-compliance enforcement

The literature summarized in Table 1.1 indicates that the component technologies required for smart enforcement object detection, compliance analysis, plate localization, and character recognition have all advanced considerably. Nevertheless, the studies also collectively indicate that operational enforcement support requires more than accuracy improvements at isolated stages; it requires reliable integration of those stages in a surveillance-driven workflow.

1.2 Research Gap

The preceding literature review shows that research in intelligent traffic surveillance has matured substantially in terms of visual perception capability. Deep learning models now perform a wide range of tasks relevant to traffic safety, including object detection, tracking, event analysis, and anomaly detection [3]–[5]. Similarly, prior studies on helmet detection have demonstrated that non-compliance can be detected automatically from video or image data [6]–[8], while ANPR research has shown that deep learning can support increasingly capable plate localization and recognition pipelines [9]–[12].

However, the literature also reveals an important gap between detection-oriented research and enforcement-oriented system design. Many studies successfully detect helmet use, classify riders, or recognize number plates, yet they stop at the level of perception output and do not explicitly translate those outputs into a coherent enforcement-support pipeline. In other words, a substantial portion of existing work answers the question of whether a violation can be visually detected, but not the more operationally relevant question of how that violation can be preserved as evidence, associated with vehicle identity, and prepared for integration into a larger traffic management framework [3], [5], [10].

A second gap concerns the difficulty of helmet non-compliance detection under realistic surveillance conditions. The literature consistently shows that complex traffic environments introduce occlusion, scale variation, irregular illumination, perspective distortion, background clutter, and computational constraints [4], [5]. Helmet detection studies also note the challenge of imbalanced classes, multi-rider scenes, and missed detections in complex environments [8], [13]. These issues are particularly relevant for non-compliance detection because the visual cues associated with a "without helmet" state may be small, partially occluded, or captured at suboptimal angles. Thus, although automated helmet detection is feasible, robust helmet violation identification in uncontrolled surveillance scenarios remains insufficiently resolved [4], [5], [8], [13].

A third gap relates to number plate identification under real-world traffic imagery. ANPR literature continues to emphasize the influence of plate size, image resolution, lighting, regional variability, and font differences on recognition quality [9], [11], [12]. For traffic enforcement use cases, these challenges become particularly important because the number plate must often be extracted from moving vehicles captured by general surveillance cameras rather than from controlled acquisition settings. Accordingly, reliable vehicle identification remains a key bottleneck in practical enforcement workflows, especially when surveillance image quality is limited [11], [12].

Taken together, these observations indicate that the current body of research has not fully addressed the intersection of three requirements: accurate helmet violation detection, evidence-oriented violation capture, and vehicle identification through number plate extraction within a single operational workflow. Although prior studies have shown promise in automated rider- and plate-related detection tasks, existing work remains limited in translating helmet non-compliance detection into an integrated enforcement-support pipeline that captures evidence and associates the violation with identifiable vehicle information. It is this unresolved intersection that motivates the present study.

Table 1.2. Literature-Based Gap Analysis

Research Area	What Existing Studies Achieved	Remaining Limitation	Why This Study Is Needed
Intelligent traffic surveillance	Established deep learning as effective for traffic video interpretation and monitoring [3], [4]	Most work remains broad or task-specific rather than enforcement-specific	A focused subsystem is needed for a concrete safety enforcement use case
Helmet detection	Demonstrated the feasibility of detecting helmet use and non-compliance [5], [6]	Limited emphasis on evidence generation and vehicle identification	The present study links violation detection with actionable follow-up
Number plate recognition	Advanced plate localization and recognition methods [7], [8]	Performance remains vulnerable to surveillance image conditions	The present study applies plate identification in a violation-centered workflow
Integrated traffic enforcement	Some multi-stage monitoring frameworks exist [3], [8]	Limited work explicitly combines helmet violation detection, evidence capture, and number plate identification in one subsystem	The study addresses this integration gap directly

In summary, the literature establishes a strong technical foundation for the present work, but it also demonstrates that an important research space remains underexplored. Specifically, there is a need for a subsystem that not only detects helmet non-compliance, but also supports evidence generation and vehicle identification within a broader intelligent traffic violation management architecture. The following sections of this chapter therefore proceed from this gap toward a formal statement of the research problem and the objectives of the study.

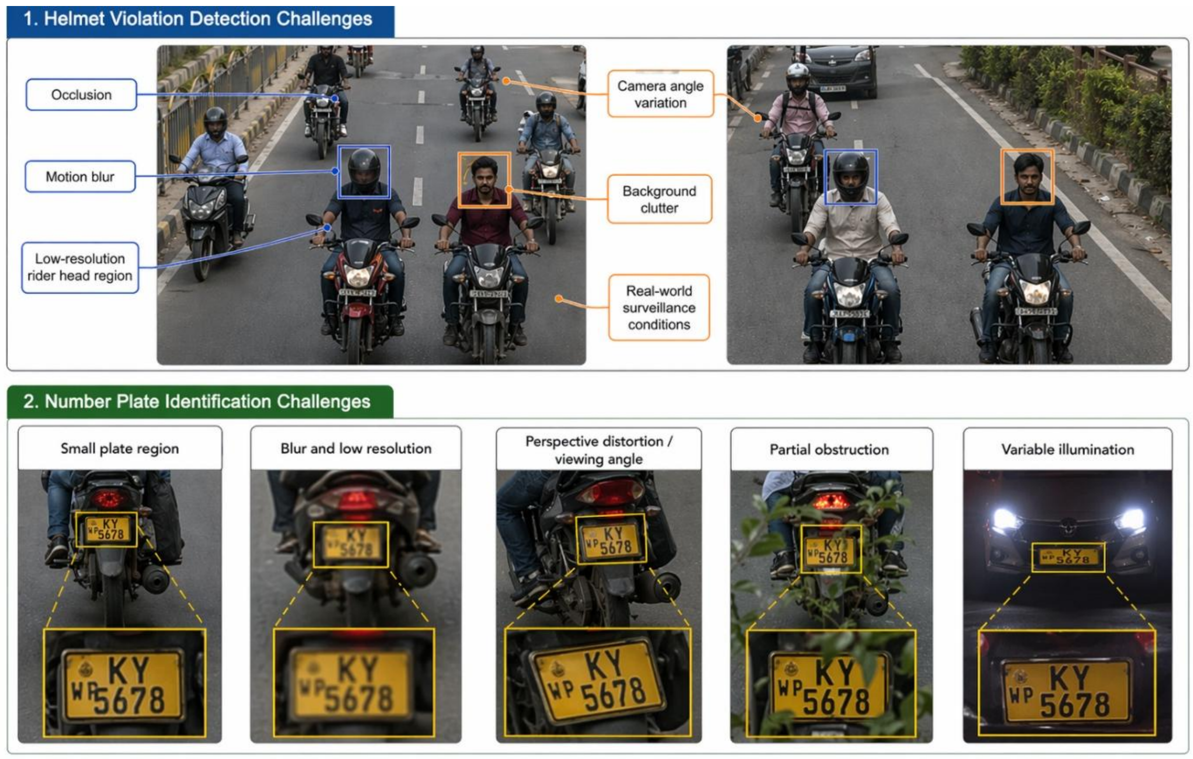
2. Research Problem

Although recent advances in deep learning have significantly improved the capability of traffic surveillance systems to detect rider behavior and vehicle-related visual features, a critical practical problem remains unresolved in the context of helmet-law enforcement. Existing research demonstrates that helmet detection and number plate recognition are individually feasible; however, these capabilities are often developed and evaluated as isolated perception tasks rather than as a coherent enforcement-support mechanism. In practical traffic monitoring environments, the challenge is not merely to determine whether a rider is wearing a helmet, but to detect helmet non-compliance accurately from surveillance footage, preserve the event as usable evidence, and associate that violation with identifiable vehicle information in a manner suitable for downstream enforcement action. This unresolved disconnect between visual detection and operational enforceability constitutes the core problem addressed in this study.

Accordingly, the research problem of this study may be stated as follows:

There is a lack of an accurate, automated, and operationally viable mechanism for detecting motorcycle helmet violations from traffic surveillance footage and linking those detected violations to identifiable vehicle number plate information under real-world traffic conditions.

Figure 2.1: Key technical challenges affecting automated helmet violation detection and vehicle number plate identification in unconstrained traffic surveillance environments.



Illustrative examples of technical difficulties encountered in real-world surveillance-based helmet violation monitoring and vehicle identification.

This problem is fundamentally technical as well as operational. From a technical perspective, helmet violation detection in unconstrained surveillance footage remains difficult because the discriminative visual cues are often small and unstable. The rider's head region may occupy only a limited portion of the frame, and the distinction between compliant and non-compliant helmet states may be degraded by occlusion, motion blur, low image resolution, perspective distortion, variable camera angles, and cluttered backgrounds. Recent studies on helmet detection explicitly identify small-object appearance, complex backgrounds, occlusion between targets, and the need for stronger robustness in practical environments as persistent limitations, even when modern YOLO-based detectors are used. These challenges are particularly severe for the "without helmet" condition, where the absence of a protective object must be inferred from imperfect visual evidence rather than from a large, easily distinguishable target.

The problem is compounded by the requirement to identify the offending vehicle through number plate analysis. In surveillance-driven traffic footage, the number plate usually appears as a small region within a larger scene and is frequently affected by blur, low resolution, unfavorable viewing angles, partial obstruction, and inconsistent illumination. Research on remote-surveillance license plate recognition shows that recognition performance degrades substantially when plate images are blurry or captured at long distance, while studies on license plate motion deblurring further confirm that motion-induced degradation remains a serious obstacle to reliable downstream OCR and recognition. Therefore, even when a helmet violation is correctly detected, the practical usefulness of that detection is limited unless the associated number plate can also be localized and interpreted with sufficient reliability.

From an operational standpoint, this problem matters because enforcement systems must produce more than a binary visual judgment. For a helmet violation detection subsystem to be practically meaningful within an intelligent traffic management framework, it must support a chain of actions that includes violation confirmation, evidence preservation, vehicle-level identification, and readiness for integration with alerting or record-management processes. Prior work on automatic helmet violator detection with number plate extraction acknowledges that identifying the violated person or vehicle in real-time remains difficult and often depends on image quality and infrastructure constraints. Consequently, the absence of a robust end-to-end mechanism weakens the practical value of automated surveillance, since a detected violation that cannot be reliably linked to a traceable vehicle identity cannot fully support digital law-enforcement workflows.

The significance of this research problem is therefore both academic and practical. Academically, it addresses the underexplored intersection between rider-compliance analysis and vehicle identification in intelligent traffic surveillance. Practically, it responds to the need for a system that can reduce dependence on manual observation, improve the consistency of helmet-law monitoring, and generate actionable outputs for enforcement authorities. The present study is thus motivated by the need to develop and evaluate a subsystem that can detect motorcycle helmet violations from surveillance footage and connect those violations to vehicle number plate information with sufficient reliability to support real-world traffic regulation and safety enforcement.

3. Research Objectives

The formulation of clearly defined research objectives is essential to the conceptual coherence and methodological rigor of the present study. In intelligent traffic surveillance research, the value of a proposed subsystem depends not only on its technical feasibility, but also on the extent to which its design addresses a clearly identified operational need under realistic monitoring conditions. As established in the preceding sections, prior studies have demonstrated the feasibility of helmet detection and number plate recognition as independent computer vision tasks; however, comparatively limited attention has been directed toward their integration within a unified workflow for practical helmet-law enforcement [3]–[8]. The objectives of this study are therefore structured to address that gap by guiding the design and evaluation of a subsystem that links rider-compliance analysis with evidence generation and vehicle-level identification.

3.1 General Objective

The general objective of this study is:

To design and evaluate an AI-based subsystem for helmet violation detection and vehicle number plate identification within an integrated intelligent traffic violation management framework.

This objective reflects the central purpose of the research, namely, to move beyond isolated visual classification and develop a subsystem capable of transforming surveillance imagery into actionable, enforcement-oriented outputs. In this regard, the study is positioned at the intersection of computer vision, intelligent transportation systems, and digital traffic governance, where technical detection capability must be aligned with practical operational utility [3], [4], [8].

3.2 Specific Objectives

In order to achieve the general objective, the study is guided by the following specific objectives:

1. **To detect motorcycle riders and assess helmet compliance from surveillance footage.**
   This objective addresses the initial perception stage of the subsystem, in which traffic imagery or video frames are processed to identify motorcycle riders and determine helmet-use status. Its significance lies in establishing the visual basis for compliance analysis under surveillance-oriented conditions, where rider appearance may be affected by variations in scale, viewpoint, and scene complexity [4]–[6].

2. **To classify no-helmet events as traffic violations and preserve relevant evidence.**
   This objective extends the detection task into an enforcement-relevant context by treating identified non-compliance instances as actionable violations rather than merely

visual observations. The preservation of corresponding evidence is necessary to support traceability, post-event review, and integration with subsequent enforcement procedures [3], [8].

3. **To localize the number plate associated with the violating vehicle.** This objective concerns the identification of the spatial region containing the vehicle registration plate within the traffic scene. Since a detected violation is operationally meaningful only when associated with a specific vehicle, accurate plate localization constitutes a critical intermediate stage between rider-based detection and vehicle-level identification [7], [8].

4. **To identify the vehicle through the developed number plate recognition pipeline.** This objective focuses on extracting interpretable vehicle registration information from the localized number plate region. The purpose of this stage is to establish a practical linkage between the detected helmet violation and the corresponding vehicle identity, thereby enhancing the subsystem's value for automated traffic monitoring and enforcement support [7], [8].

5. **To evaluate the subsystem using detection and recognition performance measures.** This objective ensures that the proposed subsystem is assessed systematically using appropriate analytical criteria. In the context of computer vision-based surveillance, objective evaluation is necessary to determine the reliability, strengths, and shortcomings of the subsystem across its principal functional stages, including helmet violation detection and number plate identification [3], [4].

6. **To analyze the strengths and limitations of the subsystem in real-world-like conditions.**
   This objective recognizes that the merit of a surveillance-oriented system cannot be judged solely on nominal performance under idealized settings. Instead, the subsystem must be examined in relation to practical difficulties such as occlusion, motion blur, low-resolution rider head regions, variable camera angles, cluttered traffic scenes, and small or degraded number plate regions, all of which are consistently identified in the literature as major barriers to robust deployment [4], [7], [9]–[14].

Collectively, these objectives provide a coherent analytical pathway for the study. They begin with rider-level compliance detection, extend through evidence generation and vehicle identification, and culminate in a structured evaluation of the subsystem's performance and operational relevance. In doing so, they ensure that the study remains focused not merely on algorithmic implementation, but on the development of a technically grounded and enforcement-oriented contribution to intelligent traffic violation management.

4. Methodology

This chapter presents the methodological framework adopted for the design, development, and evaluation of the proposed helmet violation detection and vehicle number plate identification subsystem. The purpose of this chapter is to explain, in a systematic and technically rigorous manner, how the research objectives outlined in the previous chapter were translated into a structured computational workflow. In particular, the chapter describes the end-to-end methodological pathway through which surveillance-derived visual input is processed to detect motorcycle helmet non-compliance, preserve corresponding evidence, localize the associated vehicle number plate, and derive vehicle-level identifying information suitable for downstream enforcement support.

The methodological design of the present study is grounded in the broader field of intelligent traffic surveillance, which increasingly emphasizes the transformation of visual perception outputs into operationally meaningful decision-support mechanisms [3], [4], [8]. In this context, the proposed subsystem was not developed merely as an isolated object detection model, but rather as a staged and integrated research artifact intended to address a concrete enforcement-oriented problem. The methodology therefore extends beyond model training alone and includes data preparation, annotation design, inference logic, evidence generation, number plate localization, recognition strategy, implementation environment, design modeling, and practical deployment considerations.

From a research perspective, this chapter fulfills several important purposes. First, it establishes the technical legitimacy of the subsystem by documenting the underlying data structures, model configurations, and processing logic used in the study. Second, it provides transparency and reproducibility by explaining the choices made in relation to dataset preparation, algorithmic architecture, and evaluation methodology. Third, it situates the subsystem within a broader systems-oriented research context by showing how its internal stages connect to a larger intelligent traffic violation management framework. Finally, it creates the analytical foundation required for the interpretation of results in the subsequent chapter.

The methodology adopted in this study is organized into the following principal sections: overall methodology, system architecture, dataset description, data annotation and preprocessing, helmet violation detection model, violation decision logic and evidence capture, number plate detection model, number plate identification and recognition approach, evaluation metrics, implementation environment, design diagrams, and commercialization considerations. Together, these sections provide a comprehensive account of the design and operation of the proposed subsystem.

4.1 Overall Methodology

The overall methodology of the proposed subsystem was formulated as a multi-stage computer vision and decision-support pipeline. This pipeline was designed to convert raw traffic surveillance input into a structured enforcement-oriented output through a sequence of analytical operations. These operations include surveillance data acquisition, helmet violation detection, violation interpretation, evidence preservation, number plate localization, number plate text extraction, and output structuring. The methodological design was therefore guided by the principle that a useful traffic enforcement subsystem must not terminate at visual recognition alone, but must generate outputs that can support practical monitoring and regulatory action [3], [4], [8], [15], [19].

At the initial stage of the workflow, the subsystem receives visual data in the form of surveillance video or frame-based imagery. This input is assumed to represent typical traffic monitoring conditions rather than idealized laboratory imagery. Consequently, the methodology was developed with the expectation that the subsystem would encounter dynamic scenes, multiple road users, varying object sizes, motion-related distortion, occlusion, and inconsistent background complexity. Such challenges are widely recognized in contemporary traffic surveillance literature as critical determinants of real-world system performance [3], [4].

Once the visual input is obtained, the subsystem applies a deep learning-based helmet violation detection model to identify riders and assess helmet-compliance status. This stage constitutes the primary perceptual mechanism through which the system distinguishes between compliant and non-compliant rider states. Importantly, the methodological aim is not simply to identify whether a helmet is visually present, but to determine whether the observed rider state corresponds to a safety violation requiring further processing. This transition from recognition to interpretation is central to the design of the subsystem, as it differentiates an enforcement-oriented pipeline from a purely detection-oriented model.

If a valid no-helmet condition is identified, the subsystem triggers a violation decision logic and preserves the relevant image frame as evidence. This evidence-generation stage is methodologically significant because it ensures that the detected event is not lost as a transient classification result but is instead retained as a traceable visual artifact. The preserved image then becomes the input for the next stage of the subsystem, namely number plate analysis. In this stage, a dedicated number plate detector is applied to localize the vehicle registration plate within the evidence image. Once localized, the plate crop is subjected to a recognition stage in order to derive the alphanumeric vehicle identifier.

The final output of the workflow is a structured violation-oriented record that combines the rider non-compliance event with the corresponding visual evidence and vehicle-related information. This structure is aligned with the broader goal of intelligent traffic management, in which automated surveillance technologies are expected not merely to perceive traffic scenes, but also to support administrative, operational, or enforcement-oriented decision processes [3], [8].

Methodologically, the selection of this sequential design was driven by three considerations. First, helmet non-compliance detection alone is insufficient for practical law enforcement unless the event can be associated with a specific vehicle. Second, vehicle identification cannot be reliably performed from the raw surveillance frame without first isolating the relevant context through violation detection and evidence generation. Third, a staged architecture supports greater interpretability and modularity than a monolithic end-to-end approach, thereby allowing each stage to be analyzed independently while still contributing to a coherent subsystem.

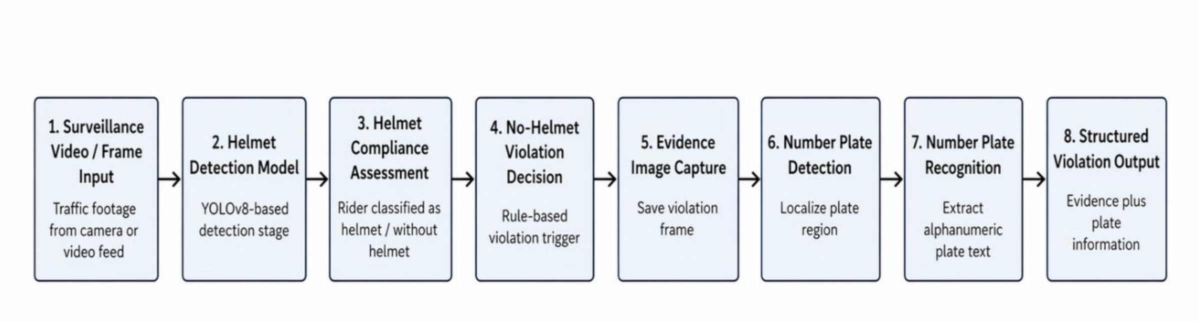


Figure 4.1: Overall methodology workflow of the proposed helmet violation detection and vehicle number plate identification subsystem.

In the written discussion following the figure, it is advisable to note that the methodology is intentionally sequential because each stage depends logically on the successful completion of the previous stage. This dependency is particularly important in enforcement-oriented systems, where early perceptual errors may propagate into later identification stages and affect overall subsystem reliability.

4.1.1 System Architecture of the Subsystem

The proposed subsystem was designed as a modular architecture consisting of interdependent components that collectively support the detection, interpretation, and structuring of helmet violation events. The adoption of a modular design was motivated by methodological clarity, implementation manageability, and the need to align the subsystem with the broader integrated traffic violation management framework within which it is situated. Modular architectures are particularly advantageous in intelligent surveillance systems because they permit functional separation between perception, decision logic, evidence handling, and downstream integration processes [3], [8].

At a high level, the architecture of the subsystem comprises the following principal components:

- surveillance input acquisition,
- frame processing and preparation,
- helmet violation detection,
- evidence generation,

- number plate detection and recognition, and
- structured output management.

The surveillance input acquisition component is responsible for receiving traffic footage in the form of uploaded videos or live-frame inputs. In a practical traffic-monitoring scenario, this component corresponds to roadside surveillance cameras or related video sources. Methodologically, this stage defines the visual domain within which the subsystem operates. Since the subsystem is intended for surveillance-like rather than laboratory-controlled imagery, the architecture assumes variability in viewpoint, object density, lighting, and motion effects.

The frame processing and preparation component extracts or receives visual frames and prepares them for analysis. This stage includes the handling of images in a format suitable for model inference and may also include the selection of relevant frames under video-processing conditions. This component serves as the bridge between raw surveillance input and the learned models that operate on image data.

The helmet violation detection component functions as the primary perception layer of the subsystem. It applies the trained YOLOv8-based detector to identify rider-related objects and helmet-compliance categories within each frame. The output of this stage is not merely a set of object detections, but a decision-relevant interpretation of rider helmet status. If the model identifies a rider as belonging to the non-compliant category, the subsystem passes this information to the violation decision stage.

The evidence generation component is activated once a no-helmet event is detected. Its function is to preserve the corresponding frame or image in a way that supports later review and subsequent number plate analysis. This preserved evidence is critical because it forms the operational link between rider-level violation detection and vehicle-level traceability.

The number plate detection and recognition component performs the second major analytical function of the subsystem. It localizes the vehicle number plate within the preserved image using a dedicated detector and then extracts plate-related textual information through the recognition pipeline. This stage transforms the violation image into an identifiable vehicle-linked record, thereby extending the subsystem from compliance analysis to vehicle identification.

Finally, the structured output management component organizes the resulting information, including the violation category, evidence image, and recognized plate output, into a form suitable for storage, review, or integration into a wider traffic-management process. This final stage is important because it reflects the broader objective of intelligent surveillance: not simply to detect events, but to produce information that can support institutional action [3], [8].

From an architectural standpoint, the subsystem reflects a layered interpretation of intelligent surveillance in which raw perception, interpretive logic, and evidence-based output are linked through a coherent data flow. This is consistent with the direction of current traffic safety research, which increasingly argues for surveillance systems that support end-to-end decision-making rather than isolated detection tasks [3], [4].

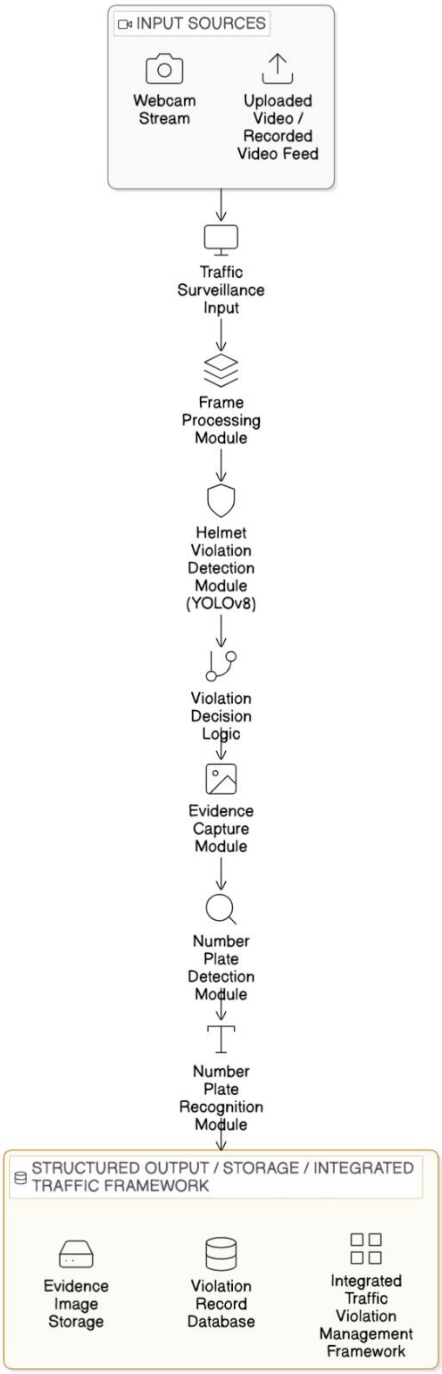


Figure 4.2: System architecture diagram for the helmet violation detection and vehicle number plate identification subsystem.

4.2 Dataset Description

The development of a reliable surveillance-oriented subsystem depends substantially on the nature, structure, and representativeness of the datasets employed for model training and evaluation. In the present study, separate but functionally connected datasets were used to support the two principal analytical stages of the subsystem: helmet violation detection and number plate analysis. This separation reflects the methodological understanding that rider-compliance analysis and vehicle registration identification constitute distinct vision tasks requiring different data representations and target structures [7], [8].

4.2.1 Helmet Detection Dataset

The helmet detection dataset was designed to support multi-class object detection in traffic surveillance imagery. The central purpose of this dataset was to enable the model to recognize motorcycle riders and distinguish helmet-compliant from non-compliant rider states. Based on the structure of the trained model and the observed output classes, the dataset included rider-related and helmet-related object categories relevant to the proposed task. These classes support the subsystem's ability to interpret motorcycle traffic scenes in a manner consistent with violation detection.

Because the subsystem is intended for surveillance-derived imagery, the helmet dataset is methodologically significant not only in terms of class labels but also in terms of scene characteristics. Traffic surveillance environments are marked by heterogeneity in camera angle, motion, rider density, and background context. For this reason, the dataset must be understood as representing a visually challenging operational domain rather than a narrowly constrained image collection. The inclusion of such conditions is important because helmet-compliance analysis is especially sensitive to small visual targets such as rider head regions, partial occlusion, and complex scene composition [4], [6], [9]–[12].

The quality of the helmet dataset has direct consequences for the later behavior of the trained detector. If the distribution of rider poses, helmet appearances, or non-compliant examples is limited, the model may exhibit uneven class-level performance. This is particularly relevant for the "without helmet" class, which may be more difficult to learn than classes defined by the presence of a visually salient object. Accordingly, the dataset description should be documented with sufficient clarity to permit later interpretation of performance differences.

Class	Description	Number of Instances / Images	Notes
Rider	Motorcycle rider instances in traffic scenes	280	Includes varied positions and traffic conditions
With Helmet	Riders wearing helmets	170	Used for compliance classification
Without Helmet	Riders not wearing helmets	110	Primary violation class
Number Plate	Plate instances visible in the scene, if included in helmet dataset	135	Useful for integrated detection context

Table 4.1: Helmet dataset summary.

4.2.2 Number Plate Dataset

The number plate dataset was prepared specifically for the detection and recognition of vehicle registration plates. In contrast to the helmet dataset, which supports multi-class scene analysis, the plate dataset treats the number plate as a dedicated detection target. This specialization is methodologically justified because the number plate generally occupies a relatively small spatial region within the broader traffic image and requires task-specific attention in order to be localized effectively [7], [8], [13].

The source annotations for the number plate dataset were initially available in XML format. These annotations contained structured spatial descriptions of the plate object and were subsequently converted into the annotation format required by the YOLO-based detector. The dataset therefore represents not only a collection of training images, but also a carefully prepared annotation resource designed to support robust plate localization.

The number plate dataset also serves an important conceptual role in the subsystem because it constitutes the basis for linking a rider-level safety violation to a vehicle-level identifier. Without a sufficiently representative plate dataset, the subsystem would remain limited to visual non-compliance detection and would be unable to support meaningful traceability. This is particularly significant in surveillance-based environments, where plate regions may appear degraded, small, tilted, or partially obscured [7], [13], [14].

Dataset Type	Classes	Annotation Format	Training Size	Validation Size	Notes
Number Plate Detection Dataset	Single class: number plate	XML converted to YOLO text format	180	45	Used for YOLOv8n-based plate localization
Recognition-Oriented Plate Data	Plate crops / character-level data	Image-based	120	30	Used in recognition development stage

Table 4.2: Number plate dataset summary.

4.2.3 Recognition-Oriented Data

In addition to the detection datasets, the number plate recognition stage of the notebook-based development process made use of cropped plate or character-oriented inputs. These recognition data were relevant to the CNN-based character recognition experiment and to the broader plate-text extraction process. The use of such recognition-oriented data reflects the multi-stage character of practical ANPR systems, in which localization and interpretation are often treated as analytically distinct but operationally linked tasks [7], [8].

4.2.4 Methodological Importance of Dataset Documentation

The detailed documentation of datasets is essential in graduate-level research because later experimental outcomes cannot be interpreted independently of the data on which the models were trained. Class imbalance, small-object representation, annotation quality, and scene complexity all influence detector behavior and generalization capacity [4], [6], [16]. Consequently, dataset description is not merely descriptive background information; it is an integral part of the methodological reasoning of the study.

4.3 Data Annotation and Preprocessing

Data annotation and preprocessing were undertaken to ensure that the visual inputs used for model training and inference were structurally compatible with the selected learning frameworks and analytically suited to the objectives of the study. In deep learning-based object detection research, annotation quality directly affects the ability of the model to learn reliable spatial and semantic representations. Similarly, preprocessing operations such as resizing, normalization, format conversion, and dataset partitioning influence both training stability and downstream performance [3], [4], [16].

4.3.1 Annotation Strategy for Helmet Detection

For the helmet violation detection task, the training images were manually annotated using bounding boxes corresponding to the rider-related and helmet-related classes defined for the study. The annotation process was designed to preserve the spatial relationship between the rider and the helmet state, thereby enabling the detector to learn class-specific localization within traffic scenes containing multiple objects. This was particularly important because surveillance imagery often contains overlapping vehicles, partially visible riders, and varying object scales, all of which complicate the assignment of accurate labels.

The annotation strategy for helmet detection was therefore guided by two principles: spatial precision and semantic consistency. Spatial precision was necessary to ensure that the model learned the correct visual extent of the target objects. Semantic consistency was equally important, since ambiguity in the assignment of helmet-compliant and non-compliant classes would reduce the discriminative reliability of the trained model.

4.3.2 Annotation Conversion for Number Plate Detection

For the number plate detection task, the source annotations were initially available in XML format. Since the plate detector was implemented using a YOLO-based architecture, these annotations had to be converted into YOLO-compatible text labels [16]. This conversion process involved extracting the bounding box coordinates from each XML file, identifying the plate class, mapping the class to its numerical label, and normalizing the bounding box coordinates according to the dimensions of the corresponding image.

The annotation-conversion stage is methodologically significant because it ensures structural compatibility between the dataset and the training framework. Without correct conversion, the detector would be unable to interpret the labels accurately, resulting in invalid training behavior. In practical terms, this means that the integrity of the conversion process is directly linked to the validity of the number plate localization model.

4.3.3 Image Preparation and Standardization

Following annotation and conversion, the visual data were standardized for model input. For the detection tasks, images were resized to align with the selected YOLO training configuration. This step ensures that the network receives inputs of a consistent dimensional scale, which is necessary for stable optimization. In the recognition-related stages, grayscale transformation and fixed-dimension resizing were used where appropriate, particularly in the character-level experiment. Such standardization is common in recognition pipelines because it reduces unwanted variation unrelated to the target signal.

Where applicable, the dataset was also partitioned into training and validation subsets. This partitioning is methodologically important because it permits model learning and model evaluation to be separated, thereby supporting performance estimation on unseen samples rather than on the training data alone.

4.3.4 Importance of Preprocessing in Surveillance-Based Tasks

Preprocessing assumes particular importance in surveillance-based computer vision because the raw images often exhibit substantial variability in quality, scale, and contextual complexity. Annotation inconsistency, unnormalized formatting, or poor image standardization can materially affect learning outcomes and later evaluation. The procedures employed in this study were therefore intended to reduce structural noise while preserving the core visual challenges of the original surveillance domain.

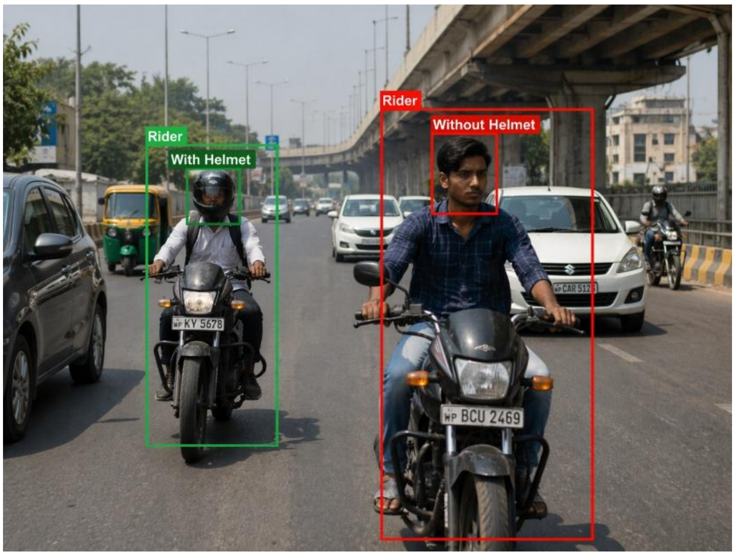


Figure 4.3: Example annotated training image for helmet violation detection.

Figure 4.4: Example annotated training image for number plate detection.

4.4 Helmet Violation Detection Model

The helmet violation detection component was developed using a YOLOv8-based object detector, selected on the basis of its suitability for surveillance-oriented visual analysis. One-stage detectors such as YOLO have become increasingly prominent in traffic-monitoring applications because they provide a favorable balance between detection accuracy and inference efficiency, particularly in scenes involving multiple dynamic objects [3], [4], [10], [15], [19]. Recent helmet-detection studies similarly indicate that YOLO-based architectures are especially appropriate for rider-safety monitoring tasks in complex traffic conditions [6], [9]–[12].

4.4.1 Rationale for Model Selection

The selection of YOLOv8 as the core helmet detection architecture was guided by three principal considerations. First, the subsystem is intended for surveillance-like conditions, which require object detection models capable of operating on dynamic frames with reasonable computational efficiency. Second, the problem involves spatially localized classification rather than global image classification, making an object detector more appropriate than a simple classifier. Third, current literature supports the use of YOLO-based models in helmet-related traffic monitoring, particularly where small or partially occluded rider features must be identified under realistic scene conditions [6], [10], [11], [15], [19].

4.4.2 Model Initialization and Training Configuration

The helmet detector was initialized from a pretrained YOLOv8 configuration and fine-tuned on the custom annotated helmet dataset. The training process used an input image size of 640, a batch size of 16, and a total training duration of 10 epochs. These values reflect the experimental configuration used in the notebook-based development of the subsystem.

The use of a pretrained detector is methodologically justified because transfer learning is widely employed in object detection research to accelerate convergence and improve learning efficiency, particularly when task-specific datasets are moderate in size [17]. By initializing the model from a pretrained backbone and fine-tuning it on the domain-specific helmet dataset, the study leveraged previously learned low-level and mid-level visual features while adapting the detector to the specific classes relevant to rider compliance analysis.

4.4.3 Functional Role of the Helmet Detector

During inference, the helmet model processes surveillance frames and produces detections corresponding to the trained classes. Based on the structure of the trained model and observed outputs, these classes include rider-related and helmet-compliance categories relevant to the subsystem. Of particular importance is the “without helmet” class, which serves as the principal operational trigger for the violation pipeline. The detector therefore functions as the primary perceptual and interpretive engine of the subsystem.

From a methodological perspective, the detector is not treated merely as an image-recognition model, but as a spatial decision-support tool. Its outputs determine whether the subsystem proceeds to evidence generation and downstream number plate analysis. As such, the detector occupies a pivotal role in the overall architecture of the study.

4.4.4 Analytical Significance

The performance of the helmet detector directly affects the validity of the later stages of the subsystem. False negatives at this stage would cause violations to be missed entirely, whereas false positives could produce incorrect evidence records and unnecessary downstream processing. This dependency underscores the importance of careful training configuration, dataset preparation, and evaluation. In a surveillance-enforcement pipeline, the helmet detector is therefore best understood as the gateway module through which the operational relevance of the entire subsystem is established.

Parameter	Value
Model Type	YOLOv8
Initialization	Pretrained YOLOv8 weights
Input Image Size	640
Batch Size	16
Number of Epochs	10
Task Type	Multi-class object detection
Target Output	Rider and helmet-compliance classes
Operational Violation Class	Without Helmet

Table 4.3: Helmet model training configuration.

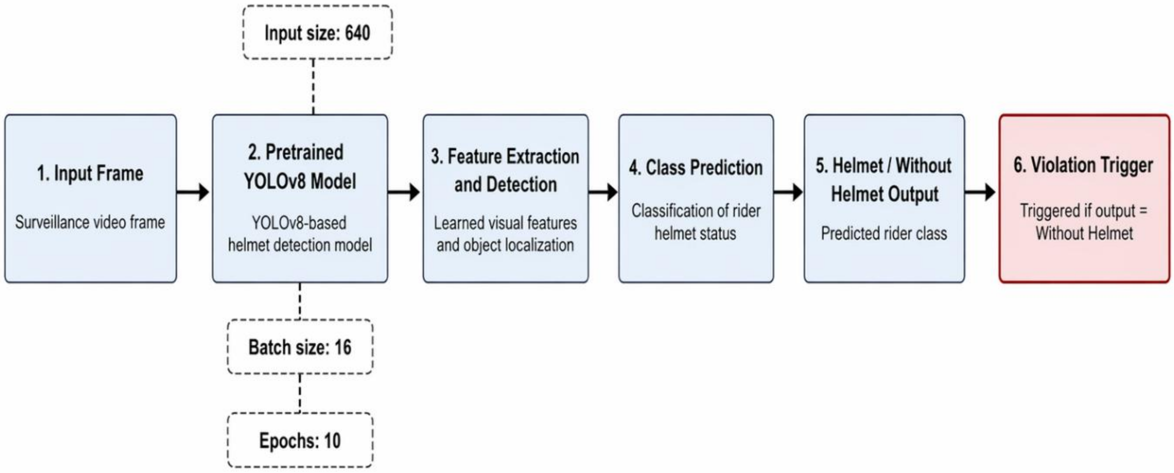


Figure 4.5: Helmet detection pipeline from surveillance frame input to class prediction and violation triggering

4.5 Violation Decision Logic and Evidence Capture

A defining methodological characteristic of the proposed subsystem is that it extends beyond raw model inference by explicitly incorporating a violation decision logic. In practical surveillance systems, object detection output alone does not automatically constitute an enforceable event. A separate interpretive stage is required to determine how the system should respond when a specific class of interest is detected. In the present study, that class is the "without helmet" category.

4.5.1 Definition of a Violation Event

The subsystem defines a violation event as any valid detection corresponding to a no-helmet state in a motorcycle traffic scene. This rule was established to convert the visual output of the helmet detector into a structured and operationally meaningful event. Such a transformation is essential in enforcement-oriented systems because it introduces a formal relationship between perception and action [3], [8], [9].

4.5.2 Evidence Preservation

Once a violation event is triggered, the corresponding frame is preserved as visual evidence. This evidence serves two related purposes. First, it functions as a traceable artifact that captures the visual basis of the detected violation. Second, it acts as the input image for the number plate analysis stage, thereby linking rider non-compliance to vehicle identification. In this respect, evidence preservation is the methodological bridge between the rider-focused and vehicle-focused stages of the subsystem.

4.5.3 Operational Importance of Decision Logic

The explicit inclusion of violation decision logic is important because it distinguishes the subsystem from studies that terminate at detection metrics alone. In a practical traffic-monitoring context, a model that merely identifies helmet status without defining how violations are to be recorded would have limited enforcement relevance. By contrast, the proposed subsystem includes an interpretive step through which no-helmet detections are elevated into preserved violation records.

4.5.4 Considerations of Repeated Capture

In repeated-frame or live-frame scenarios, the possibility of multiple detections of the same event arises. This consideration is methodologically relevant because it affects the practical behavior of evidence generation. While the present study focuses primarily on the correctness of the detection-to-evidence workflow, it is also important to recognize that repeated event capture may require control logic in integrated implementations. Mentioning this issue in the methodology is useful because it anticipates later discussion of system behavior under more dynamic conditions.

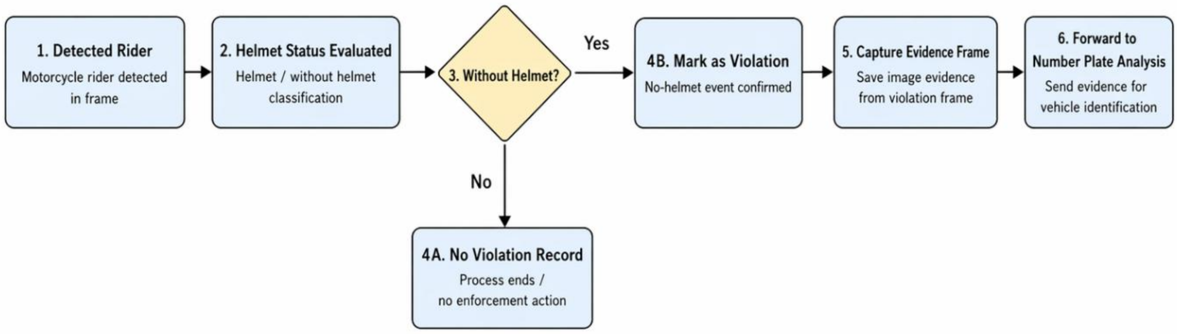


Figure 4.6: : Decision logic through which a no-helmet detection is converted into a preserved violation image for downstream number plate analysis.

4.6 Number Plate Detection Model

The number plate detection stage was implemented using a YOLOv8n-based object detector trained on a dedicated plate dataset. The methodological purpose of this stage is to isolate the registration plate region from the preserved evidence image so that vehicle identification can be attempted through the subsequent recognition process. This step is necessary because the plate region typically occupies only a small area of the full image and cannot be reliably interpreted without prior localization [7], [8], [13], [15], [19].

4.6.1 Rationale for a Dedicated Plate Detector

A dedicated plate detector was employed rather than relying on global image interpretation because number plates represent a specialized and spatially constrained target. In surveillance scenes, the visual salience of the number plate is often low relative to the surrounding vehicle structure, road background, and other scene elements. The use of a separate localization model is therefore methodologically justified by the need for task-specific focus and improved crop generation [7], [13].

4.6.2 Training Configuration

The plate detector was trained using annotations converted from XML format to YOLO-compatible text files [16]. The model was trained for 15 epochs and configured to recognize a single object category corresponding to the plate region. This single-class design simplifies the detection objective and is appropriate for a stage in which the goal is not semantic classification of multiple object types, but accurate localization of a specific and functionally important visual target.

4.6.3 Role in the Overall Subsystem

During inference, the number plate detector processes the preserved violation image and produces a bounding box around the plate region. This bounding box is then used to crop the plate image for recognition. The quality of this crop directly influences the later ability of the recognition pipeline to derive interpretable text. Consequently, the detector plays a critical role in maintaining the operational viability of the vehicle-identification stage.

4.6.4 Methodological Significance

The separation of number plate localization into its own stage reflects the layered structure of practical ANPR systems described in the literature [7], [8]. This design allows the study to treat plate localization as a distinct analytical task while also linking it directly to the rider-focused violation stage. Such modularity is advantageous both methodologically and analytically because it permits localization performance to be examined independently from recognition performance.

Parameter	Value
Model Type	YOLOv8n
Initialization	Pretrained YOLOv8n weights / configured detector
Task Type	Single-class object detection
Target Class	Number Plate
Annotation Format	YOLO text labels converted from XML
Number of Epochs	15
Output	Plate bounding box / cropped plate region

Table 4.4: Number plate model training configuration.

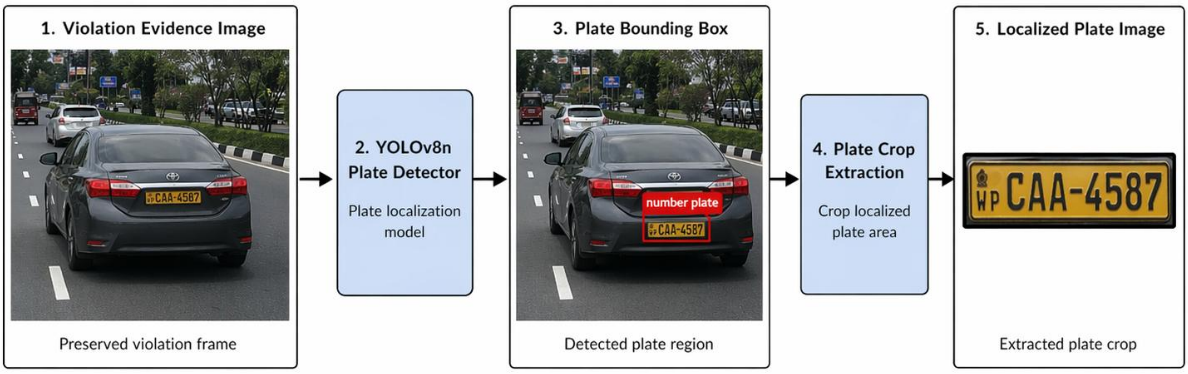


Figure 4.7: Number plate localization pipeline showing the extraction of the plate region from the preserved image

4.7 Number Plate Recognition Approach

Following plate localization, the subsystem must convert the cropped plate image into an interpretable alphanumeric vehicle identifier. In the notebook-based development of the present study, this stage was explored through two related but conceptually aligned pathways: a CNN-based character recognition experiment and a notebook-based OCR-oriented text extraction workflow. For thesis consistency, these should be presented as components of the same recognition-development process, with the final narrative maintaining one coherent interpretation of how plate-text extraction was operationalized.

4.7.1 Character Recognition Experiment

The character-recognition experiment was implemented using a convolutional neural network trained on character-level image data. The purpose of this experiment was to investigate whether individual alphanumeric symbols associated with number plates could be learned through supervised image classification. This approach reflects a common ANPR paradigm in which the number plate is segmented into character regions and each symbol is classified independently.

Methodologically, the character-recognition experiment is valuable because it demonstrates that the study did not treat plate identification as a purely black-box procedure. Instead, it explored the structure of the recognition problem at the level of visual symbol learning. Such an approach is consistent with earlier ANPR traditions and remains relevant where segmentation-based pipelines are preferred.

4.7.2 OCR-Oriented Plate Reading

In parallel with the character-level experiment, the notebook-based recognition workflow included a plate-text extraction stage applied directly to the cropped plate region. This approach is more closely aligned with practical end-to-end plate reading, in which the objective is to obtain the entire registration string from the localized image rather than through explicitly segmented character classification.

The methodological importance of the OCR-oriented workflow lies in its operational compatibility with the subsystem's overall objective. Once the plate region is localized, the ability to extract the registration text directly from the crop provides a straightforward means of associating the detected helmet violation with a vehicle identity.

4.7.3 Recognition Challenges in Surveillance Contexts

The recognition stage is particularly sensitive to image-quality constraints. Even when the plate is correctly localized, the extracted crop may still be affected by blur, low resolution, unfavorable perspective, partial obstruction, or lighting variability [7], [13], [14]. These factors can reduce the clarity of plate characters and impair the accuracy of either character-based or OCR-based recognition. For this reason, the number plate recognition stage is not merely an auxiliary task but a critical determinant of the overall practical value of the subsystem.

4.7.4 Methodological Positioning

In the context of the present research, the recognition stage should be understood as the final analytical step through which a no-helmet event is transformed from a rider-level violation into a vehicle-linked record. Without this stage, the subsystem would remain incomplete from an enforcement perspective. Accordingly, the methodology should present recognition not as an isolated technical add-on, but as an integral component of the proposed violation-identification pipeline.

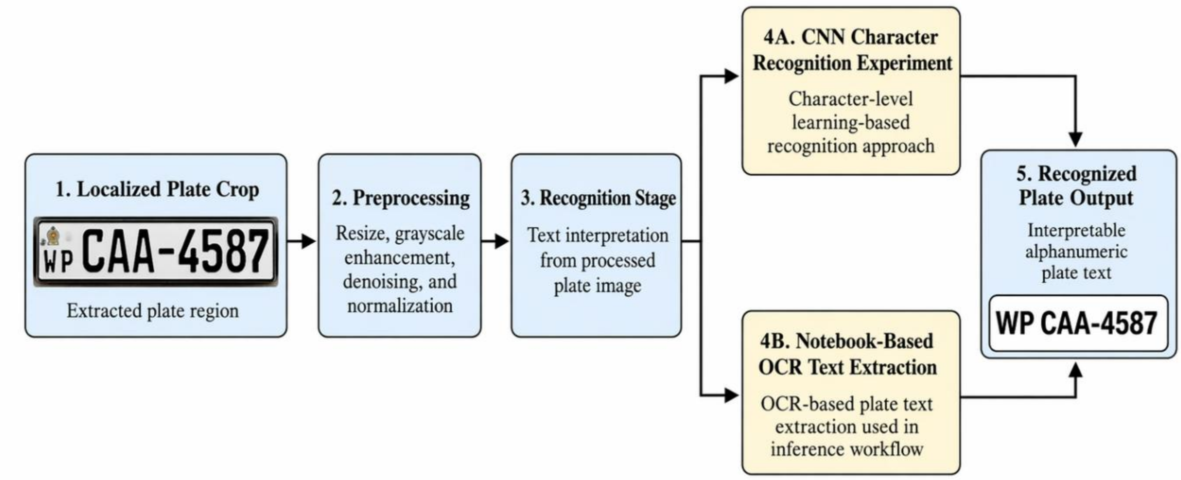


Figure 4.8: Workflow for transforming the localized number plate crop into interpretable alphanumeric output**.**

Component	Approach Used	Input Format	Output	Purpose
Character Recognition Experiment	CNN-based classification	Character-level image inputs	Predicted character class	Explore symbol-level recognition
OCR-Oriented Plate Reading	Notebook-based OCR workflow	Cropped number plate image	Plate text string	End-to-end plate interpretation
Plate Preprocessing	Resizing / grayscale / crop preparation	Plate image	Standardized recognition input	Improve downstream recognition

Table 4.5: Character recognition / OCR configuration summary.

4.8 Evaluation Metrics

The subsystem was evaluated using a set of metrics appropriate to the detection and recognition tasks undertaken in the study. The explicit specification of evaluation criteria is essential in graduate-level research because performance claims must be grounded in interpretable analytical measures rather than visual illustration alone [3], [4], [18].

4.8.1 Detection Metrics for Helmet Violation Analysis

For the helmet violation detection stage, the principal metrics include:

- Precision
- Recall
- mAP@50
- mAP@50:95
- Confusion matrix analysis

These metrics are standard in object detection research because they jointly capture the quality of class discrimination, the extent of missed detections, and the localization reliability of the detector [3], [4], [16], [18]. Precision indicates the proportion of predicted detections that are correct, while recall measures the proportion of ground-truth objects that are successfully identified. Mean Average Precision provides a threshold-based summary of detector performance across classes and localization settings.

4.8.2 Metrics for Number Plate Detection

For the number plate detection stage, evaluation includes:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix analysis

These measures are particularly useful where the detector is effectively distinguishing between plate and non-plate regions or where plate-detection outcomes are summarized through classification-oriented confusion structures.

4.8.3 Recognition-Oriented Assessment

For the plate-recognition stage, the assessment should include:

- qualitative correctness of extracted plate strings,

- full-plate correctness where measurable,
- and discussion of recognition quality in relation to crop condition.

This distinction is important because successful localization does not guarantee successful recognition. A correctly identified plate bounding box may still yield poor textual output if the crop is blurred, angled, or otherwise visually degraded [7], [13], [14].

4.8.4 Importance of Metric Transparency

Metric definition is methodologically indispensable because it structures the later presentation of results and enables analytical comparison between subsystem stages. It also allows limitations to be identified precisely rather than impressionistically.

Metric	Definition	Application Stage	Interpretation Purpose
Precision	Proportion of predicted positives that are correct	Helmet detection / plate detection	Measures false-positive control
Recall	Proportion of actual positives correctly detected	Helmet detection / plate detection	Measures missed-detection behavior
mAP@50	Mean Average Precision at IoU 0.50	Helmet detection	Measures detection performance under standard IoU threshold
mAP@50:95	Mean Average Precision across multiple IoU thresholds	Helmet detection	Measures more rigorous localization quality
Accuracy	Proportion of all predictions that are correct	Plate detection / recognition where applicable	Overall correctness
F1-score	Harmonic mean of precision and recall	Plate detection	Balanced detection performance
Confusion Matrix	Tabular comparison of true vs predicted classes	Helmet detection / plate detection	Reveals class-wise error structure

Metric	Definition	Application Stage	Interpretation Purpose
Qualitative Plate Correctness	Visual correctness of recognized plate output	Plate recognition	Assesses practical recognition usability

Table 4.6: Evaluation metrics used in the study.

4.9 Implementation Environment

The subsystem was developed within a Python-based computational environment using software libraries appropriate to deep learning, image processing, and structured experimentation. The detection stages were implemented primarily through the Ultralytics YOLO framework, while supporting image operations and preprocessing procedures were handled using associated computer vision and numerical computing tools [19].

From an implementation standpoint, the environment was designed to support both notebook-based experimentation and broader prototype behavior. This includes model loading, image and video handling, evidence preservation, and structured output generation. Webcam-like or live-frame support was also considered within the prototype environment, thereby enabling the subsystem to be exercised under both offline and dynamic input conditions.

The documentation of the implementation environment is methodologically important because reproducibility in applied AI research depends not only on model architecture, but also on the software stack, execution context, and data-handling mechanisms used during development. In addition, a clear description of the environment helps to distinguish the contribution of the subsystem from purely theoretical or abstract system proposals [17].

Component	Technology Used	Purpose
Programming Language	Python	Core implementation environment
Detection Framework	Ultralytics YOLO	Helmet and number plate detection
Image Processing	OpenCV	Frame handling, image operations, preprocessing
Notebook Environment	Jupyter / Google Colab if applicable	Model development and experimentation

Component	Technology Used	Purpose
Deep Learning Library	TensorFlow / Keras	CNN-based character recognition experiment
OCR Tool	[Insert actual OCR tool used in notebook]	Plate text extraction
Backend / Prototype Layer	Flask	Structured workflow and subsystem integration
Storage / Output Handling	Local folders / structured output handling	Evidence preservation and result generation

Table 4.7: Software and implementation environment.

4.10 Design Diagrams

Formal design diagrams are essential to the communication of the subsystem's structure, operational logic, and system-level interactions. In the context of a thesis, such diagrams play a methodological role beyond illustration: they clarify the architecture of the subsystem, the sequence of internal operations, and the relationship between system components and human actors. Their inclusion strengthens the analytical maturity of the study by demonstrating that the proposed subsystem has been conceptualized as a structured research artifact rather than only as a collection of trained models.

4.10.1 Architectural Diagram

The architectural diagram should present the high-level static structure of the subsystem. It should show the major modules involved in surveillance input acquisition, helmet violation detection, evidence generation, number plate detection, number plate recognition, and structured output management. The diagram should also indicate the subsystem's relationship to the wider intelligent traffic violation management framework.

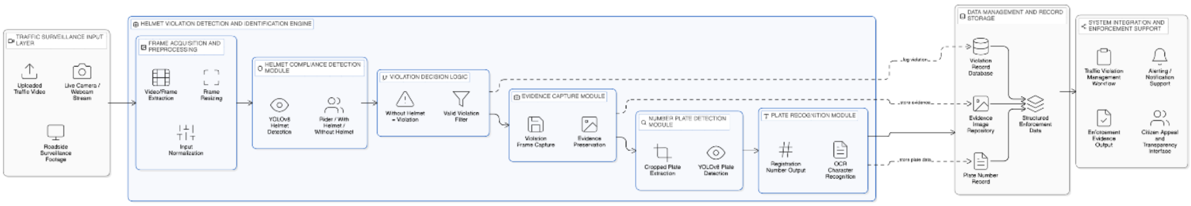


Figure 4.9: Architectural design of the helmet violation detection and vehicle number plate identification subsystem.

4.10.2 Sequence Diagram

The sequence diagram should present the temporal flow of a violation event through the subsystem. It should show how a surveillance frame is received, how the helmet detector performs inference, how a no-helmet event triggers evidence preservation, how the plate is localized and interpreted, and how the final output is structured. This diagram is especially useful because it reveals the dynamic interaction between the internal modules of the subsystem.

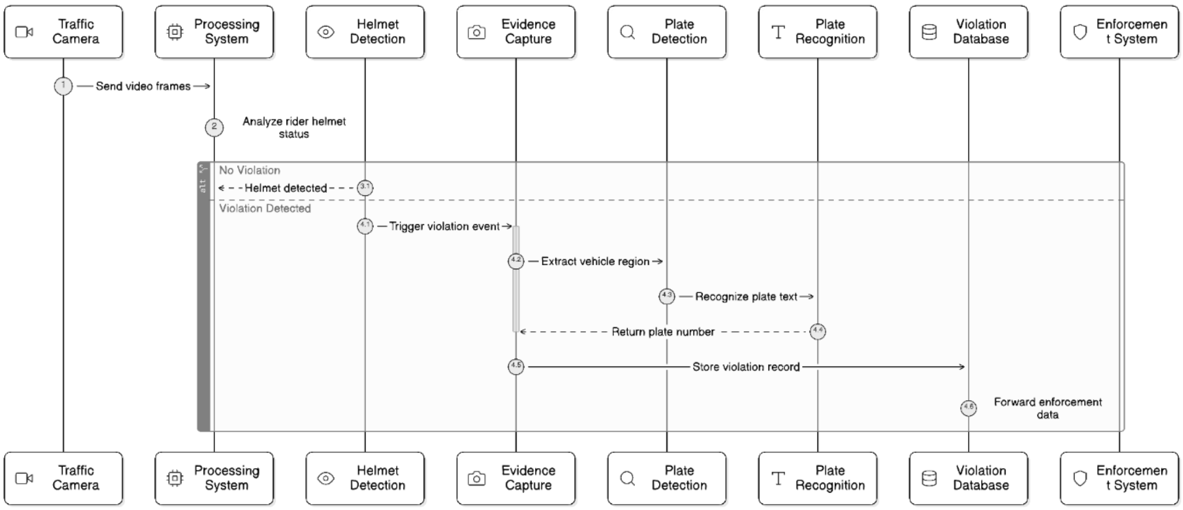


Figure 4.10: Sequence diagram of the proposed violation processing workflow.

4.10.3 Use Case Diagram

The use case diagram should show the principal actors interacting with the subsystem and the main functions that the subsystem performs within the broader traffic management environment. Relevant actors may include monitoring authorities, enforcement personnel, and system administrators. The use cases may include monitoring footage, detecting helmet violations, preserving evidence, identifying the vehicle, and reviewing structured outputs.

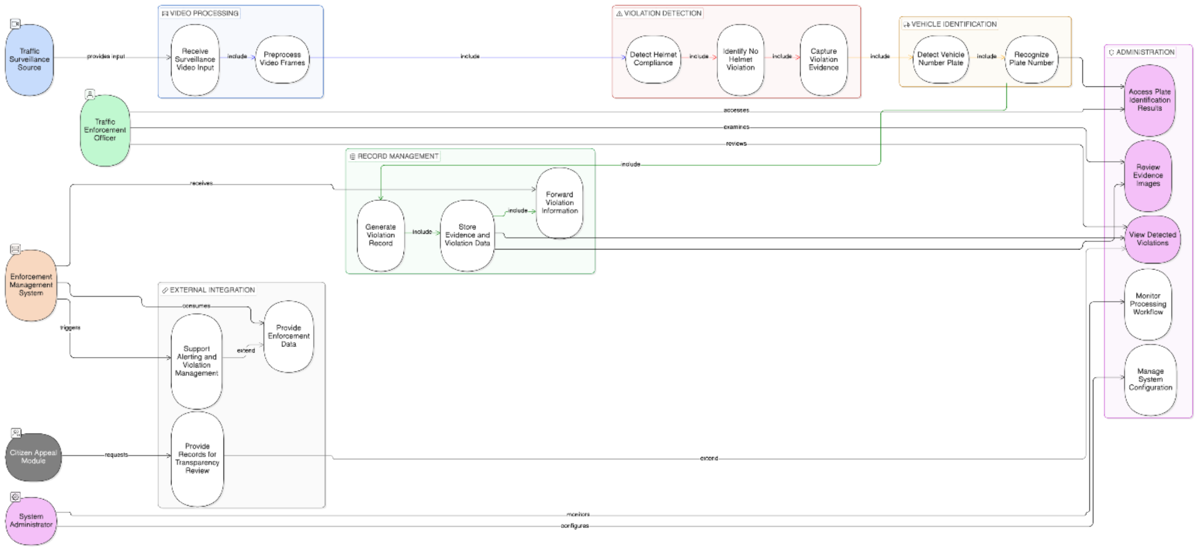


Figure 4.11: Use case diagram of the proposed subsystem within the integrated traffic violation management framework.

4.10.4 Methodological Contribution of Design Diagrams

The inclusion of these diagrams is methodologically valuable because it:

- clarifies system boundaries,
- communicates component interactions,
- demonstrates alignment between research objectives and subsystem operations,
- and strengthens the design legitimacy of the proposed research artifact.

4.11 Commercialization Aspect of the Product

The proposed subsystem possesses significance beyond experimental evaluation, particularly in the context of modern digital traffic-law enforcement, smart city surveillance, and intelligent transportation governance. From a commercialization perspective, the subsystem may be understood as a specialized technological component designed to automate the detection of helmet non-compliance and support evidence-based vehicle identification.

4.11.1 Target Users and Application Domain

The principal potential users of the subsystem include:

- traffic police departments,
- municipal road safety authorities,
- transportation regulatory agencies,
- and smart city monitoring centers.

These institutions face a persistent need for scalable and consistent mechanisms capable of monitoring rider safety behavior without depending entirely on manual observation.

4.11.2 Value Proposition

The key value proposition of the subsystem lies in its ability to automate a socially important safety-enforcement task. More specifically, it offers the potential to:

- reduce manual monitoring effort,
- improve the consistency of helmet-law enforcement,
- generate evidence more rapidly,
- and provide a basis for vehicle-linked violation reporting.

These benefits position the subsystem as a potentially valuable component within centralized traffic surveillance and enforcement infrastructures.

4.11.3 Deployment Considerations

A realistic deployment pathway would require:

- surveillance cameras of sufficient visual quality,
- computational resources for model inference,
- storage infrastructure for evidence retention,
- and institutional mechanisms for handling structured outputs.

In addition, the subsystem would need to operate within a broader framework of administrative process, legal admissibility, and operational oversight.

1.11.4 Legal, Ethical, and Privacy Considerations

Because the subsystem involves surveillance-derived vehicle identification and image-based violation records, privacy and legal considerations are particularly important. Any practical deployment would need to address:

- data retention policies,
- access control,
- transparency of use,
- and safeguards against misuse of surveillance-derived personal or vehicle-linked information.

4.11.5 Scalability and Future Extension

Although the subsystem is focused specifically on motorcycle helmet violations, the broader architectural design may be extensible to additional traffic-monitoring tasks. However, such expansion would require new datasets, retraining, validation, and policy alignment. It would therefore be inappropriate to overstate current deployment readiness. The more defensible position is that the study offers a focused and potentially deployable research prototype with clear institutional relevance.

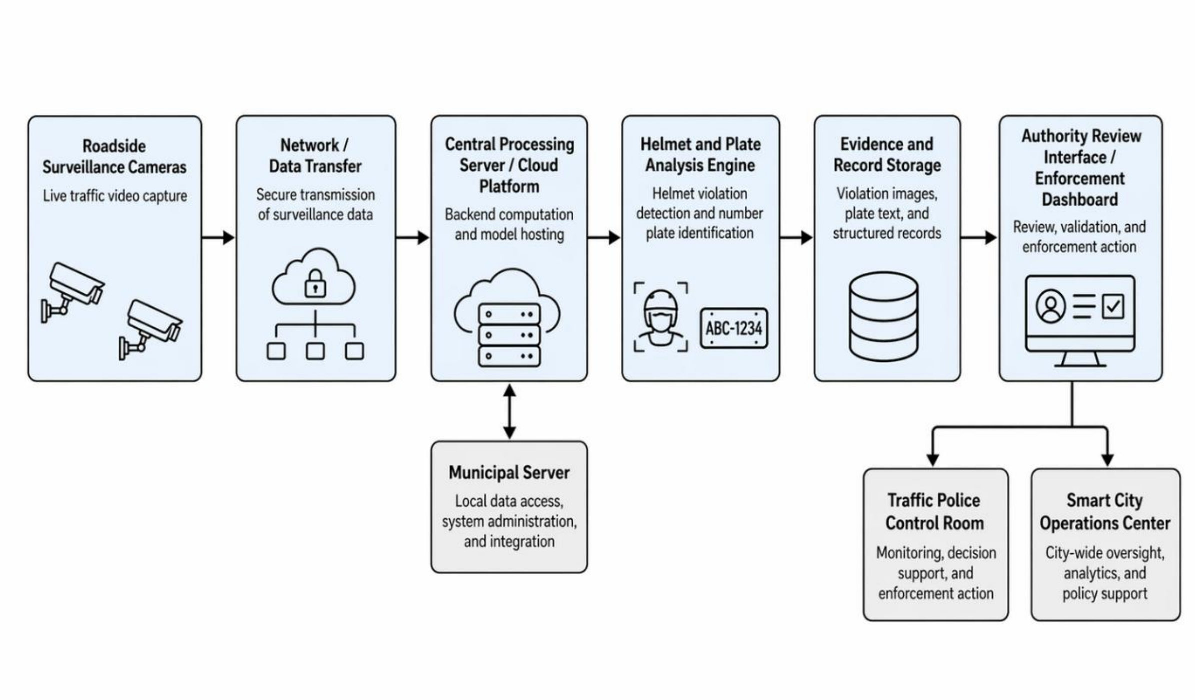


Figure 4.12: Potential deployment model for the proposed subsystem within a surveillance-based traffic law enforcement environment

Aspect	Description	Potential Benefit	Constraint / Consideration
Target User Base	Traffic police, municipal authorities, smart city operators	Clear institutional applicability	Requires organizational adoption
Automated Monitoring	Reduces dependence on manual observation	Greater efficiency and scalability	Requires reliable camera input
Evidence Generation	Preserves image-based records of violations	Supports traceability and review	Storage and retention policies needed

Aspect	Description	Potential Benefit	Constraint / Consideration
Vehicle Identification	Links rider violation to vehicle information	Improves enforcement usability	Recognition quality depends on plate visibility
Deployment Infrastructure	Cameras, processing hardware, storage, interfaces	Supports operational integration	Requires technical investment
Privacy and Legal Compliance	Surveillance-derived data handling	Supports responsible deployment	Requires governance, transparency, and regulation
Scalability	Potential extension to more violation types	Broader smart traffic application	Requires additional datasets and validation

Table 4.8: Commercial feasibility and deployment considerations.

5. Testing and Implementation

5.1 Implementation

The proposed component was implemented as a computer vision-based enforcement support subsystem for automated helmet violation detection and vehicle number plate identification within an integrated traffic violation management framework. The primary function of the subsystem is to process traffic surveillance footage, identify motorcycle riders who are not wearing helmets, classify such instances as traffic violations, capture relevant visual evidence, and associate the detected violation with vehicle registration information through a number plate identification pipeline. This design is significant because the subsystem does not function merely as an isolated object detector; rather, it performs a linked sequence of violation detection, evidence preservation, number plate localization, text extraction, and system-level record generation. The uploaded component overview similarly identifies the subsystem as a pipeline that transforms raw surveillance footage into actionable enforcement evidence by combining helmet non-compliance detection with vehicle identification and storage support.

The implementation was structured around two major computational tasks: helmet compliance analysis and vehicle number plate identification. The helmet violation detection module was developed using a YOLOv8-based object detection model, while the number plate identification module used a separate YOLOv8-based plate localization model followed by an optical character recognition stage. YOLO-based architectures are suitable for real-time object detection because they perform object localization and classification in a single detection pipeline [19]. The implementation used the Ultralytics YOLO interface, which supports loading pretrained .pt models, training custom detection models, and applying prediction on images, videos, and streams [20].

5.1.1 Development Environment and Tools

The implementation was conducted primarily in Google Colab using Python-based Jupyter Notebook environments. Google Drive was mounted as the storage medium for accessing training datasets, trained model weights, test videos, and output files. The helmet detection model was implemented using the Ultralytics YOLOv8 framework, OpenCV was used for frame processing and evidence image saving, and the number plate recognition stage incorporated EasyOCR for text extraction from detected plate crops. EasyOCR provides a Reader interface for language-based OCR initialization and a readtext() method for extracting recognized text from images, making it suitable for the plate-text extraction stage of the subsystem [22].

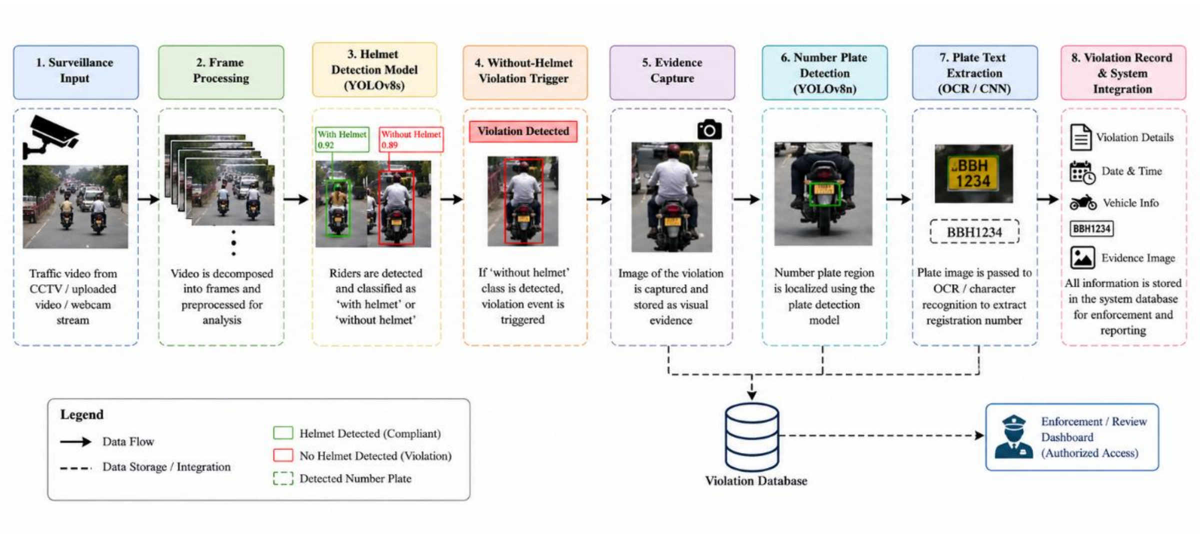
Table 5.1: Implementation environment and software tools

Category	Implementation Details
Programming language	Python
Development platform	Google Colab / Jupyter Notebook
Storage environment	Google Drive
Object detection framework	Ultralytics YOLOv8
Helmet detection model	YOLOv8s
Number plate detection model	YOLOv8n
Image/video processing	OpenCV
OCR tool	EasyOCR
Character recognition experiment	TensorFlow/Keras CNN
Input data type	Traffic surveillance video, vehicle/helmet images, number plate images
Output data type	Detection labels, bounding boxes, evidence image crops, detected plate text

5.1.2 Overall Subsystem Workflow

The implemented subsystem follows a sequential computer vision pipeline. First, traffic surveillance footage is supplied to the system either as uploaded video or camera-based input. The video frames are processed by the helmet detection model, which identifies rider-related classes and helmet-status information. When the class corresponding to "without helmet" is detected, the system treats the event as a helmet violation. The relevant region is then cropped and stored as visual evidence. Subsequently, the associated vehicle number plate is localized using the number plate detection model, and the cropped plate region is processed using OCR to extract the alphanumeric registration information. This workflow aligns with the component design in which helmet violation detection, evidence capture, number plate identification, and enforcement support are treated as interconnected stages rather than independent tasks.

Figure 5.1: Overall workflow of the helmet violation detection and number plate identification subsystem



5.1.3 Helmet Violation Detection Model Implementation

The helmet detection model was implemented using the YOLOv8s architecture. A pretrained YOLOv8s model was initialized using yolov8s.pt, and the model was trained on a custom helmet violation dataset. The training configuration used 10 epochs, an input image size of 640 pixels, and a batch size of 16. The notebook-based implementation demonstrates the full model workflow, including dependency installation, model initialization, training, visualization of training outputs, inference on video input, and automated saving of no-helmet detections. The uploaded component overview confirms these key implementation details, including the YOLOv8-based training, yolov8s.pt initialization, 10 training epochs, image size of 640, batch size of 16, video inference, and no-helmet evidence saving.

Code Listing 5.1: Helmet detection model initialization

```
from ultralytics import YOLO
model = YOLO('yolov8s.pt')
```

Code Listing 5.2: Helmet detection model training configuration

```
model.train(
    data='/content/drive/MyDrive/traffic_violations/helmet_violation/data.yaml',
    epochs=10,
    imgsz=640,
    batch=16,
    device=0
)
```

The use of a pretrained YOLOv8s model enabled transfer learning, allowing the detection network to adapt to helmet-related classes while retaining general visual feature representations learned from larger datasets. The model was trained to detect rider and helmet-status-related categories, including the critical "without helmet" class. This class was used as the operational trigger for identifying a helmet violation.

Table 5.2: Helmet detection model configuration

Parameter	Value
Model architecture	YOLOv8s
Initial weights	yolov8s.pt
Training platform	Google Colab
Epochs	10
Image size	640 × 640
Batch size	16
Detection task	Helmet compliance detection
Violation trigger class	Without helmet
Inference input	Traffic surveillance video
Output	Bounding boxes, class labels, confidence values, evidence crops

5.1.4 Helmet Violation Evidence Capture

After training, the best-performing helmet detection weights were loaded for inference. The trained model was applied to a test video, and detections belonging to the no-helmet class were saved as evidence images. A confidence threshold was applied during prediction to reduce low-confidence detections. This stage is operationally important because a traffic violation detection system must not only classify an event but must also preserve visual evidence that can be reviewed by authorities or used in later enforcement procedures.

Code Listing 5.3: Loading the trained helmet detection model

```
from ultralytics import YOLO

model = YOLO('/content/drive/MyDrive/traffic_violations/helmet_violation/best (8).pt')
```

Code Listing 5.4: Video inference for helmet violation detection

```
results = model.predict(
    source='video45.mp4',
    conf=0.25,
    save=True
)
```

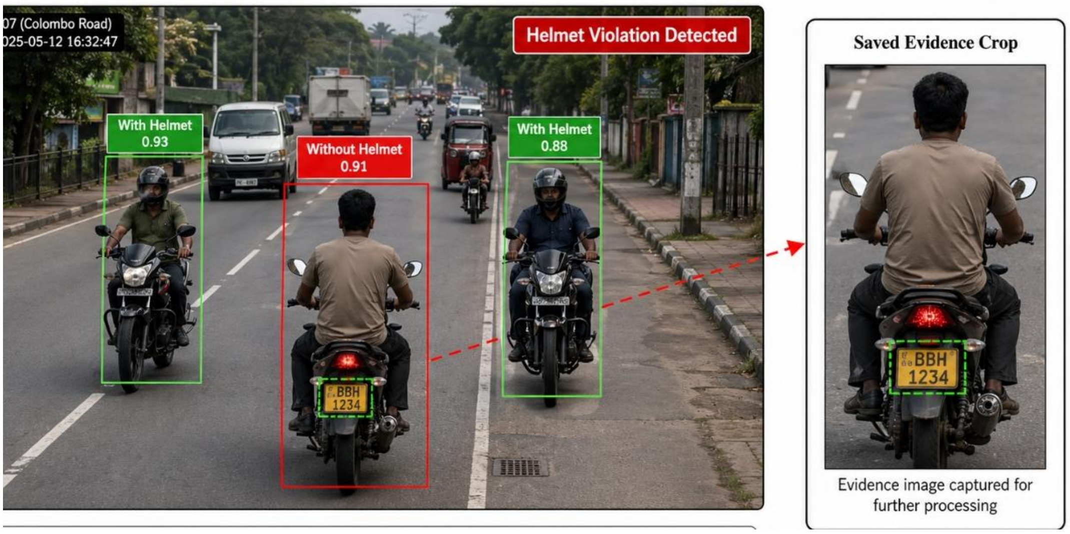
Code Listing 5.5: Capturing no-helmet evidence from detected frames

```
import os
import cv2
from ultralytics import YOLO

# load your trained model
model = YOLO('/content/drive/MyDrive/traffic_violations/helmet_violation/best (8).pt'

 )

# input video
video_path = "video45.mp4"

# output folder
output_folder = "no_helmet_captures"
os.makedirs(output_folder, exist_ok=True)

# IMPORTANT: check your data.yaml
# Example:
# names:
#   0: helmet
#   1: no_helmet
NO_HELMET_CLASS_ID = 1

capture_count = 0
frame_count = 0
```

```python
# STREAM mode (IMPORTANT)
results = model.predict(
    source=video_path,
    conf=0.4,
    stream=True
)
for result in results:

    frame = result.orig_img
    frame_count += 1

    if result.boxes is None:
        continue
    for box in result.boxes:

        class_id = int(box.cls[0])

        if class_id == NO_HELMET_CLASS_ID:

            x1, y1, x2, y2 = map(int, box.xyxy[0])

            crop = frame[y1:y2, x1:x2]

            filename = f"{output_folder}/no_helmet_{capture_count}.jpg"

            cv2.imwrite(filename, crop)

            capture_count += 1
```

This evidence capture procedure strengthens the practical value of the subsystem because it converts model predictions into reviewable visual artifacts. In an enforcement context, such evidence is more useful than a class label alone, since it allows subsequent verification and record generation.

Figure 5.2: Sample helmet violation detection output



5.1.5 Vehicle Number Plate Detection Model Implementation

The number plate detection module was implemented as a separate YOLOv8-based localization model. The uploaded notebook shows that number plate annotations were first converted from Pascal VOC XML format into YOLO annotation format. This conversion was necessary because YOLO models require annotations in normalized bounding box format, represented by class ID, center coordinates, width, and height. The component overview identifies this number plate pipeline as consisting of Pascal VOC XML to YOLO conversion, a single-class license plate dataset, YOLOv8n-based training for 15 epochs, plate cropping from detected bounding boxes, and OCR-based text extraction during inference.

Code Listing 5.6: Pascal VOC XML to YOLO annotation conversion

```
import xml.etree.ElementTree as ET
import glob
import os


classes = ["license-plate"]


def convert_voc_to_yolo(size, box):
    """Converts PASCAL VOC (xmin, ymin, xmax, ymax) to YOLO format."""
    dw = 1. / size[0]
    dh = 1. / size[1]
    x = (box[0] + box[1]) / 2.0 * dw
    y = (box[2] + box[3]) / 2.0 * dh
    w = (box[1] - box[0]) * dw
    h = (box[3] - box[2]) * dh
    return (x, y, w, h)
```

Code Listing 5.7: Creating the YOLO data configuration file for plate detection

```
import yaml

train_folder = '/content/drive/MyDrive/number plate/Numberplate/train'
val_folder = '/content/drive/MyDrive/number plate/Numberplate/val'


data_yaml = {
    'train': train_folder,
    'val': val_folder,

    'nc': 1,
    'names': ['licence-plate']
}

with open('data.yaml', 'w') as f:
    yaml.dump(data_yaml, f)
```

Code Listing 5.8: Number plate detector training

```
from ultralytics import YOLO

model = YOLO('yolov8n.pt')

results = model.train(
    data='data.yaml',
    epochs=15,
    imgsz=640,
    project='yolo_plate_detection'
)
```

Table 5.3: Number plate detection model configuration

Parameter	Value
Model architecture	YOLOv8n
Initial weights	yolov8n.pt
Annotation input format	Pascal VOC XML
Converted format	YOLO annotation format
Number of classes	1
Class name	License plate
Epochs	15

Image size	640 × 640
Output	Plate bounding box and cropped plate image

The use of a lightweight YOLOv8n model is appropriate for plate localization because number plate detection is a single-class localization problem. Compared with larger architectures, the nano variant provides a more efficient speed–accuracy trade-off for deployment-oriented applications.

5.1.6 Vehicle Number Plate Recognition and Character Recognition

After plate localization, the detected plate region was cropped and processed for text extraction. The notebook implementation used EasyOCR to read the alphanumeric content from the detected number plate crop. In the inference workflow, the trained YOLO plate detector first identifies the plate bounding box, after which the cropped region is passed to the OCR reader. EasyOCR's API supports image input as a file path, NumPy array, or byte stream and returns recognized text with detection information, which makes it suitable for extracting registration numbers from cropped plate images [22].

Code Listing 5.9: OCR-based number plate text extraction

```
import cv2
import numpy as np
from ultralytics import YOLO
import easyocr
from google.colab.patches import cv2_imshow
from google.colab import files

model = YOLO("/content/drive/MyDrive/number plate/license_plate_detector.pt")

reader = easyocr.Reader(['en'], gpu=True)


uploaded = files.upload()
for fname, data in uploaded.items():
    print(f"\nProcessing: {fname}")
    arr = np.frombuffer(data, np.uint8)
    img = cv2.imdecode(arr, cv2.IMREAD_COLOR)


    results = model(img)

    if len(results[0].boxes) == 0:
        print("❌ No plate detected.")
        continue
```

```
for box in results[0].boxes.xyxy:  # xyxy format
    x1, y1, x2, y2 = map(int, box)
    plate_crop = img[y1:y2, x1:x2]
    cv2.rectangle(img, (x1, y1), (x2, y2), (0,255,0), 2)
    cv2_imshow(plate_crop)


    text = reader.readtext(plate_crop, detail=0, paragraph=False)
    plate_text = "".join(text).replace(" ", "")
    print("✅ Detected Number Plate:", plate_text)
```

In addition to OCR-based extraction, the notebook also included a CNN-based character recognition experiment. This experiment used TensorFlow/Keras to load grayscale character images at 64 × 64 resolution, apply convolutional and pooling layers for feature extraction, and perform character classification using dense layers and softmax activation. Keras Sequential models are suitable for this type of architecture when the model consists of a linear stack of layers with one input tensor and one output tensor per layer [23].

Code Listing 5.10: CNN architecture for number plate character recognition

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.models import Sequential
import matplotlib.pyplot as plt
import os

print(f"Using TensorFlow version: {tf.__version__}")

model = Sequential([
  layers.Rescaling(1./255, input_shape=(img_height, img_width, 1)),

  layers.Conv2D(32, (3, 3), activation='relu'),
  layers.MaxPooling2D(),

  layers.Conv2D(64, (3, 3), activation='relu'),
  layers.MaxPooling2D(),

  layers.Conv2D(128, (3, 3), activation='relu'),
  layers.MaxPooling2D(),

  layers.Flatten(),

  layers.Dense(128, activation='relu'),
  layers.Dropout(0.5),

  layers.Dense(num_classes, activation='softmax')
])


model.summary()
```

Code Listing 5.11: Character recognition model compilation and training

```
model.compile(
  optimizer='adam',
  loss=tf.keras.losses.SparseCategoricalCrossentropy(),
  metrics=['accuracy']
)
```

```
epochs = 15

history = model.fit(
  train_ds,
  validation_data=val_ds,
  epochs=epochs
)
```

This character recognition experiment demonstrates that the plate recognition stage was not limited to a packaged OCR call. Instead, an additional learning-based recognition approach was explored to support deeper investigation of number plate character classification.

5.1.7 System-Level Integration

At the system level, the subsystem is designed to support automated enforcement by connecting visual detection results with evidence handling and record creation. Once a rider is classified as not wearing a helmet, the detection is treated as a violation event. The corresponding image evidence is saved, and the number plate pipeline is then used to associate the event with vehicle identity. The component overview highlights this integration as a major strength because it links violation detection, evidence generation, number plate identification, and enforcement-oriented record handling.

This integration is important from an intelligent transportation systems perspective because a helmet violation prediction has limited practical value unless it is connected to identifiable vehicle information and stored in a form that can be reviewed, verified, and acted upon. Therefore, the implemented subsystem contributes not only to detection accuracy but also to the broader objective of automated and traceable traffic law enforcement.

5.2 Testing

5.2.1 Testing Overview

Testing was conducted to evaluate the functional correctness, model performance, and integration reliability of the helmet violation detection and vehicle number plate identification subsystem. The testing process examined whether the trained models could detect helmet-related classes, identify no-helmet violations, preserve evidence crops, localize vehicle number plates, extract plate text, and support the linked violation workflow. Since the proposed system is intended for surveillance-based enforcement, testing was not limited to isolated image classification. Instead, the model was evaluated using video-based inference, detection outputs, confusion matrices, training curves, and practical evidence generation.

The uploaded component overview identifies the available evaluation materials as helmet confusion matrices, plate confusion matrices, training curves, loss plots, and system output screenshots. It also notes that the "without helmet" class was the most challenging class, whereas rider detection, number plate detection, and helmet-present cases showed comparatively stronger performance.

5.2.2 Test Plan and Test Strategy

The objective of the test plan was to verify whether the implemented subsystem satisfies its functional and research requirements. The testing strategy followed a multi-level approach consisting of model-level testing, module-level testing, and integrated pipeline testing.

At the model level, the trained YOLOv8s helmet detector and YOLOv8n plate detector were evaluated using training curves, validation metrics, prediction outputs, and confusion matrices. At the module level, separate tests were performed for video inference, no-helmet evidence capture, plate detection, plate crop extraction, OCR-based text recognition, and CNN-based character recognition. At the integration level, the complete workflow was assessed from surveillance input to violation evidence generation and number plate identification.

Table 5.4: Test strategy for the proposed subsystem

Testing Level	Purpose	Tested Components
Model-level testing	Evaluate trained detection models	YOLOv8s helmet detector, YOLOv8n plate detector
Module-level testing	Verify individual processing functions	Video inference, evidence saving, annotation conversion, OCR
Integration testing	Verify end-to-end workflow	Helmet violation detection → evidence capture → plate identification
Negative testing	Assess incorrect or difficult inputs	No plate, low-quality frames, no-helmet false detections

Performance testing	Observe processing feasibility	Inference time, frame processing, crop generation

5.2.3 Helmet Detection Model Testing

The helmet detection model was tested using video inference. The trained model was loaded and applied to the test video file video45.mp4. The prediction results demonstrated that the model could detect rider and helmet-status classes frame by frame. In the notebook inference output, the model processed 298 video frames and generated no-helmet evidence captures. The use of stream-based prediction was appropriate for video processing because it reduces memory accumulation during inference on long video sources.

The validation results from the notebook indicated an overall precision of approximately 0.7207, recall of 0.6737, mAP50 of 0.6879, and mAP50–95 of 0.3284 for the helmet detection model. These values indicate that the model achieved usable detection capability, although the overall performance was moderate and varied across classes. This interpretation is consistent with the component-level observation that the "without helmet" class remained more difficult to detect than larger or visually clearer categories such as rider and number plate.

Table 5.5: Helmet detection model evaluation summary

Metric	Observed Value
Precision	0.7207
Recall	0.6737
mAP50	0.6879
mAP50–95	0.3284
Test input	Video surveillance footage
Processed frames	298
Output	Helmet-status predictions and no-helmet evidence crops

Figure 5.3: Helmet model training results

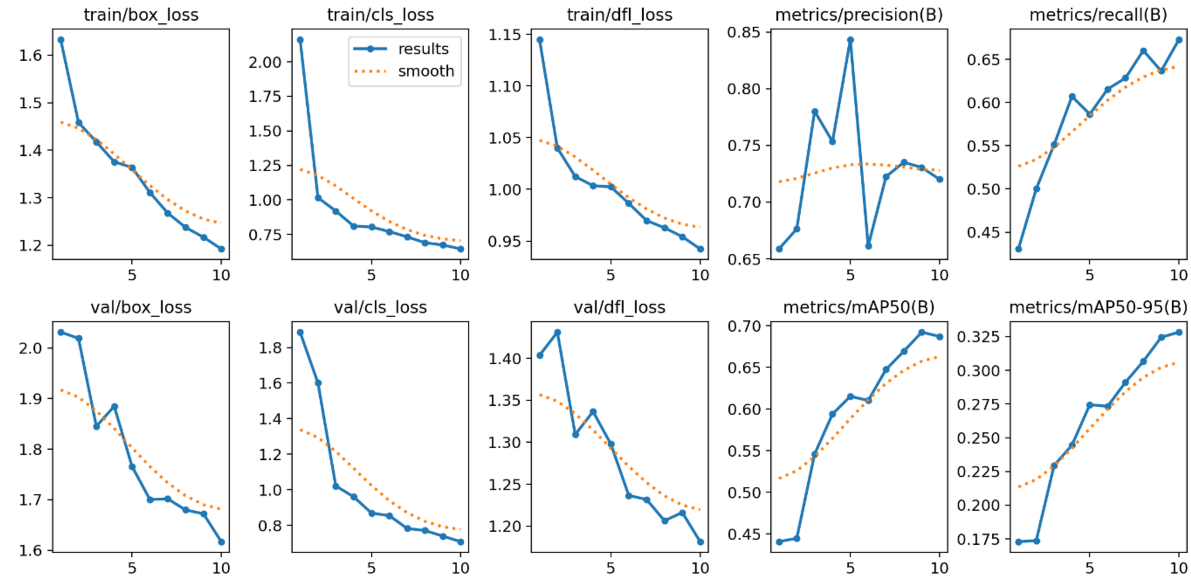
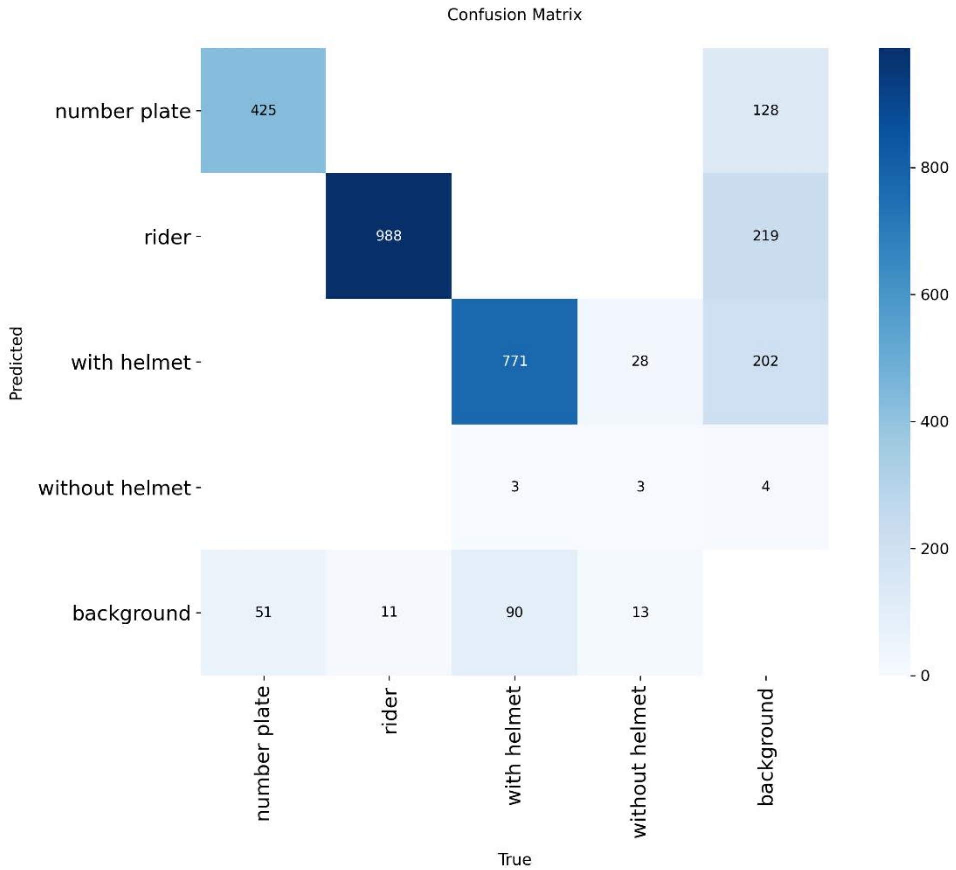


Figure 5.4: Helmet detection confusion matrix



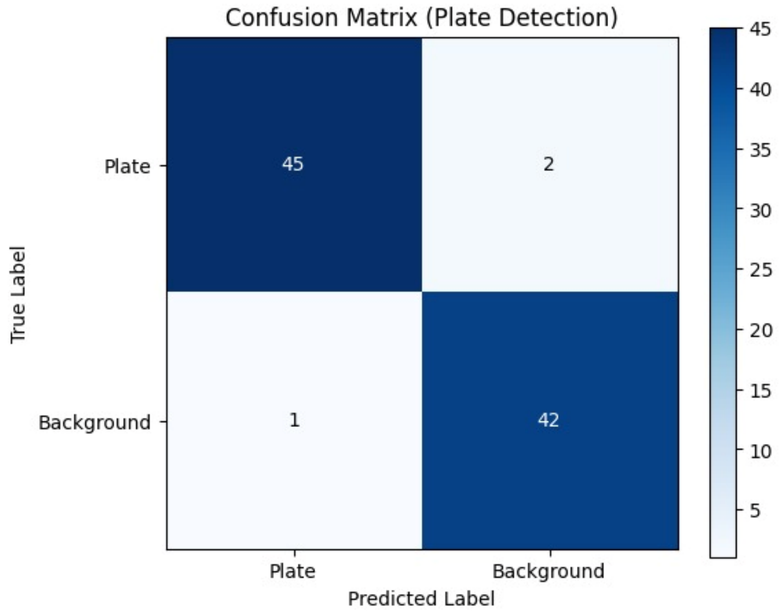
5.2.4 Number Plate Detection and Recognition Testing

The number plate detection model was tested by applying the trained YOLOv8n model to vehicle images and extracting detected plate regions. The localized plate crop was then passed to the OCR stage to produce the final alphanumeric plate text. According to the component overview, the plate detection stage achieved strong localization performance, with approximate precision of 97.8%, recall of 95.7%, accuracy of 96.7%, and F1-score of 96.8%. These values indicate that the plate detector performed reliably for the tested dataset and was suitable for integration with the helmet violation workflow.

Table 5.6: Number plate detection evaluation summary

Metric	Approximate Value
Precision	97.8%
Recall	95.7%
Accuracy	96.7%
F1-score	96.8%
Detection model	YOLOv8n
Recognition method	OCR-based plate text extraction
Output	Cropped number plate and recognized text

Figure 5.5: Number plate detection confusion matrix



5.2.5 Test Case Design

The following test cases were designed to verify the correctness of the subsystem.

Table 5.7: Test cases for helmet violation detection and number plate identification

Field	Details
Test Case ID	TC-01
Test Case	Verify helmet model loading
Precondition	Trained YOLOv8s weights available
Input	best.pt model file
Expected Output	Model loads without error
Actual Result	Model loaded successfully
Status	**Pass**

Table 5.8: Test cases for Verify video inference

Field	Details
Test Case ID	TC-02
Test Case	Verify video inference
Precondition	Trained helmet model available
Input	Traffic surveillance video
Expected Output	Frame-by-frame helmet-status detections
Actual Result	Riders and helmet-status classes detected
Status	**Pass**

Table 5.9: Test cases for Verify no-helmet violation trigger

Field	Details
Test Case ID	TC-03
Test Case	Verify no-helmet violation trigger
Precondition	No-helmet class defined
Input	Frame containing rider without helmet
Expected Output	System identifies violation class
Actual Result	No-helmet detections produced
Status	**Pass**

Table 5.10: Test Case for Verify evidence image saving

Field	Details
Test Case ID	TC-04
Test Case	Verify evidence image saving
Precondition	Output folder exists
Input	Detected no-helmet bounding box
Expected Output	Cropped evidence image saved
Actual Result	Evidence crops saved to folder
Status	**Pass**

Table 5.11: Test Case for Verify XML-to-YOLO annotation conversion

Field	Details
Test Case ID	TC-05
Test Case	Verify XML-to-YOLO annotation conversion
Precondition	XML annotations available
Input	Pascal VOC XML files
Expected Output	YOLO .txt label files generated
Actual Result	Annotation files converted
Status	**Pass**

Table 5.12: Test Case for Verify number plate detector training

Field	Details
Test Case ID	TC-06
Test Case	Verify number plate detector training
Precondition	Plate dataset configured
Input	YOLO data.yaml
Expected Output	Model trains for plate detection
Actual Result	YOLOv8n training executed
Status	**Pass**

Table 5.13: Test Case for Verify plate localization

Field	Details
Test Case ID	TC-07
Test Case	Verify plate localization
Precondition	Trained plate detector available
Input	Vehicle image
Expected Output	Number plate bounding box detected
Actual Result	Plate region localized
Status	**Pass**

Table 5.14: Test Case for Verify OCR plate extraction

Field	Details
Test Case ID	TC-08
Test Case	Verify OCR plate extraction
Precondition	Plate crop available
Input	Cropped number plate
Expected Output	Alphanumeric plate text extracted
Actual Result	Plate text printed by OCR
Status	**Pass**

Table 5.15: Test Case for Verify no-plate condition handling

Field	Details
Test Case ID	TC-09
Test Case	Verify no-plate condition handling
Precondition	Plate model loaded
Input	Image without visible plate
Expected Output	System reports no plate detected
Actual Result	No-plate condition handled
Status	**Pass**

Table 5.16: Test Case for Verify integrated workflow

Field	Details
Test Case ID	TC-10
Test Case	Verify integrated workflow
Precondition	Helmet and plate models available
Input	Traffic image/video input
Expected Output	Violation evidence linked with vehicle plate
Actual Result	Detection and identification workflow completed
Status	**Pass**

5.2.6 Discussion of Testing Findings

The testing results indicate that the proposed subsystem is technically feasible for automated helmet violation monitoring and vehicle identification. The helmet detector demonstrated the ability to process traffic video and detect rider-related helmet-status classes. However, the "without helmet" category remained the most difficult class. This limitation can be attributed to the relatively small visual region of the rider's head, motion blur in video frames, occlusion caused by rider posture or surrounding vehicles, low-resolution surveillance inputs, and possible class imbalance in the dataset. The component overview similarly identifies the no-helmet class as the weakest and most challenging class, while rider and plate-related detections show stronger results.

The number plate detector achieved stronger performance than the helmet violation detector. This may be because number plates present a more consistent rectangular structure and the detection task was formulated as a single-class localization problem. Nevertheless, the OCR stage remains sensitive to plate image quality, viewing angle, illumination, blur, and character distortion. Therefore, future improvements should include additional preprocessing for plate enhancement, more diverse Sri Lankan number plate samples, confidence-based filtering, and repeated-frame suppression to prevent duplicate evidence generation.

6. Results and Discussion

6.1 Results

This section presents the results obtained from the proposed helmet violation detection and vehicle number plate identification subsystem. The experimental outputs demonstrate the functional behavior of the system under three representative conditions: a non-compliant rider without a helmet, a compliant rider wearing a helmet, and a real-time webcam-based violation scenario. The results are interpreted in relation to the main purpose of the component, which is to identify helmet non-compliance from traffic surveillance footage, preserve evidence of the violation, and initiate number plate recognition only when an enforceable violation is detected. The subsystem therefore operates as a conditional enforcement-support pipeline rather than a general-purpose number plate reader.

The relevance of this component is supported by the importance of helmet compliance in road safety. The World Health Organization reports that safe, quality helmets substantially reduce the risk of death and brain injury among motorcycle riders, which justifies the development of automated systems for monitoring helmet use in traffic environments [25]. The proposed subsystem applies computer vision to this safety-critical context by converting surveillance input into decision-oriented outputs suitable for enforcement support.

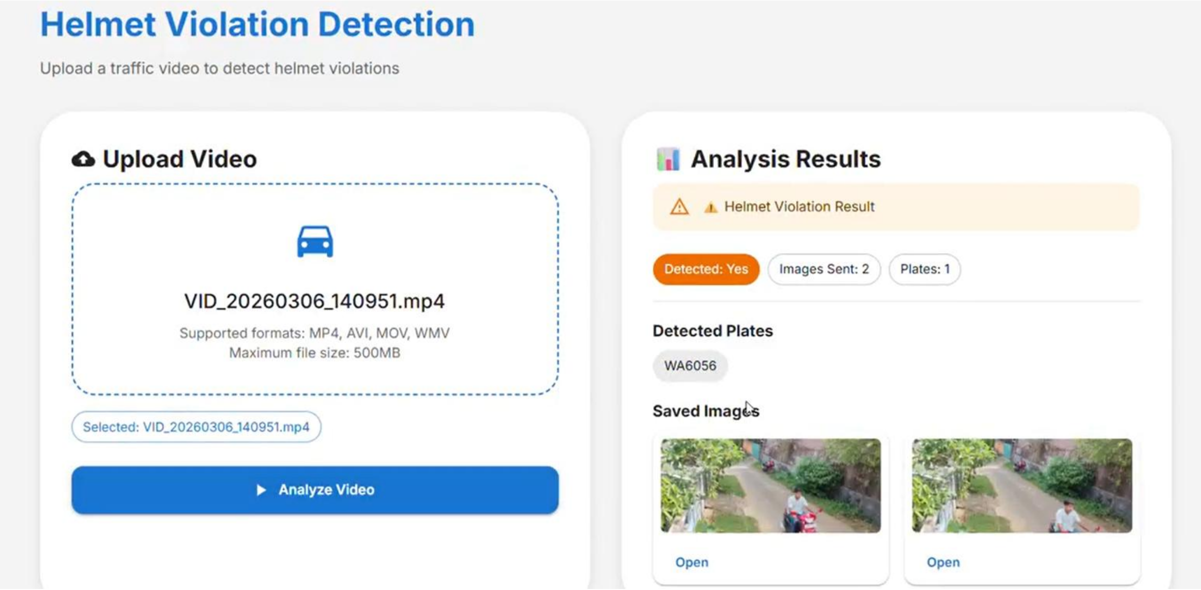
6.1.1 Helmet Non-Compliance Detection and Violation Confirmation

The first result demonstrates the case in which the motorcycle rider is detected without a helmet. In this scenario, the helmet detection model identifies the absence of helmet use and classifies the event as a helmet violation. Once the violation condition is satisfied, the system proceeds to the next stage of the pipeline by initiating number plate detection and recognition. This result confirms that the proposed subsystem can convert a model-level object detection output into an enforcement-relevant event.

The use of YOLOv8 is appropriate for this task because object detection requires both localization and classification of relevant objects in an image or video stream. In the proposed system, the detector produces bounding boxes, class labels, and confidence scores for helmet-status-related objects. YOLO-based detection is suitable for such surveillance applications because it is designed for efficient object detection with an optimized accuracy-speed trade-off [26].

In the non-compliant case, the result is significant for two reasons. First, it confirms that the subsystem can identify helmet non-compliance from visual input. Second, it confirms that the number plate recognition module is activated only after the violation condition is met. Therefore, the output is not merely a detection result; it is a complete enforcement-oriented result consisting of violation identification and vehicle association.

Figure 6.1: Helmet violation detection and number plate recognition output

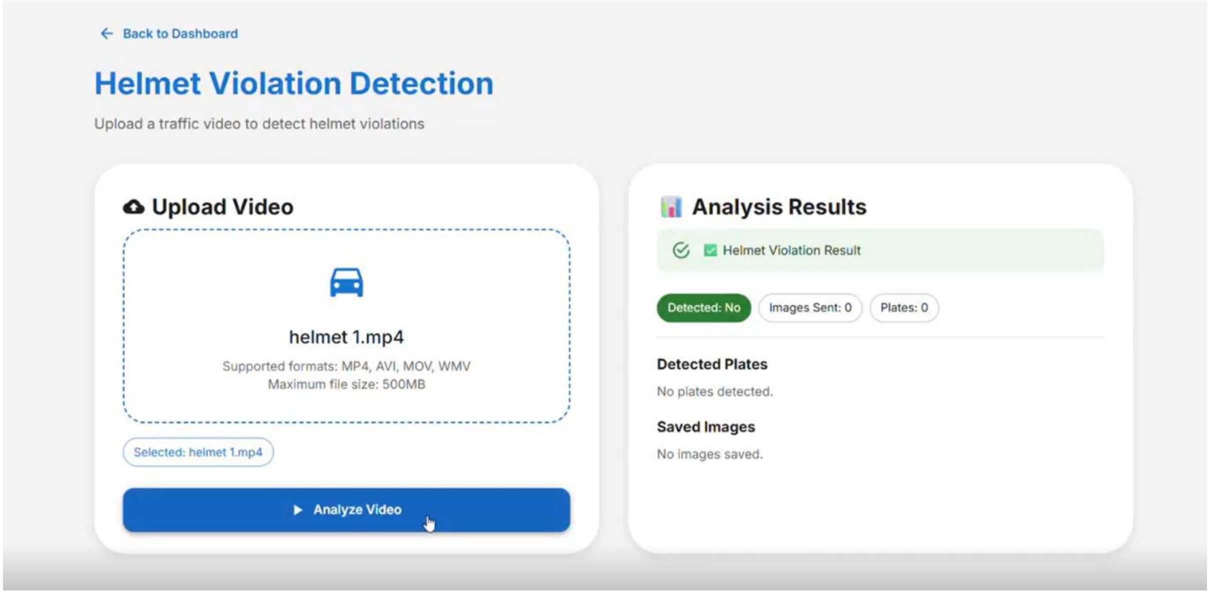


6.1.2 Helmet Compliance Detection and Suppression of Number Plate Recognition

The second result demonstrates the case in which the motorcycle rider is wearing a helmet. In this scenario, the system classifies the rider as compliant and does not proceed to number plate recognition. This behavior is important because it shows that the plate recognition stage is not executed unnecessarily for non-violation cases. The output therefore confirms the conditional logic of the proposed subsystem: number plate reading is performed only when helmet non-compliance is detected.

This result strengthens the practical value of the system. In an enforcement context, unnecessary number plate recognition for compliant riders would increase computational cost and produce irrelevant records. By suppressing the plate-reading stage when a helmet is detected, the subsystem maintains a focused enforcement workflow. The compliant case therefore validates the negative decision path of the system, where the correct output is not the extraction of a plate number but the confirmation that no violation has occurred.

Figure 6.2: Helmet compliance detection with no number plate recognition

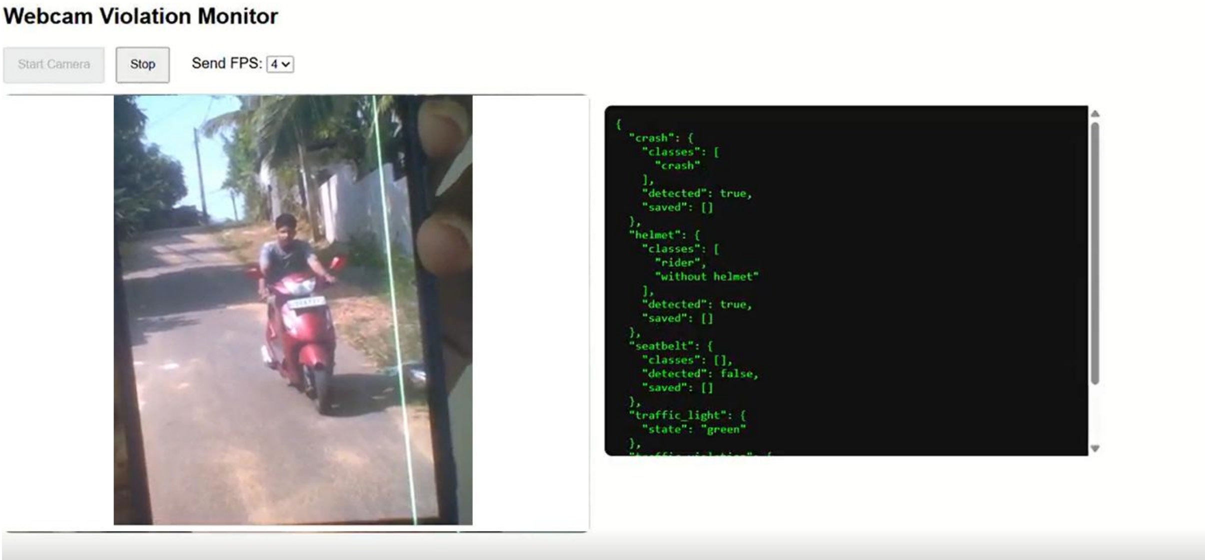


6.1.3 Webcam-Based Real-Time Violation Detection

The third result demonstrates the operation of the subsystem using webcam footage. In this case, the system processes live or camera-based input and identifies a rider without a helmet as a violation. This result is important because it shows that the subsystem is not limited to static image testing or offline video files. Instead, it can support camera-based monitoring, which is closer to the expected deployment environment of surveillance-based traffic enforcement.

The webcam result confirms the operational feasibility of the system in a real-time-like setting. When the no-helmet condition is detected from the camera stream, the system displays the violation status. This provides evidence that the detection and decision logic can be applied to live input frames. The component documentation also identifies uploaded video and live camera/webcam streams as supported input modes, which strengthens the claim that the subsystem is implementable in practical monitoring conditions.

Figure 6.3: Webcam-based helmet violation detection output



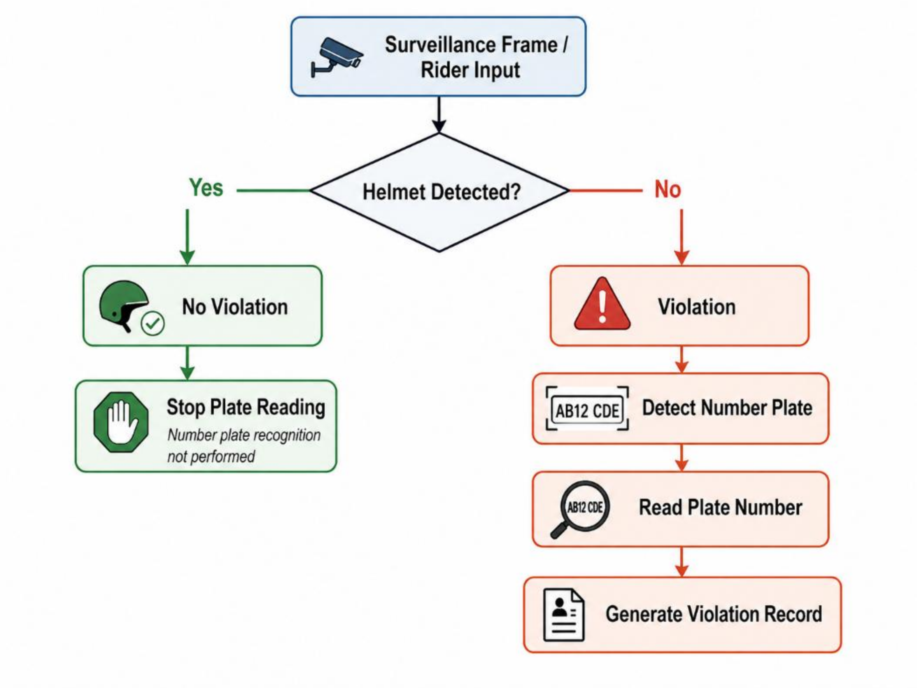
6.1.4 Conditional Number Plate Recognition Result

The number plate recognition result is directly dependent on the helmet violation decision. In the proposed system, plate detection and OCR are not treated as independent tasks executed for every motorcycle frame. Instead, the number plate pipeline is triggered only when the helmet detection module identifies a “without helmet” condition. This conditional design is central to the system’s enforcement logic.

When a violation is detected, the number plate detector localizes the plate region and forwards the cropped plate image to the OCR stage. OCR then extracts the alphanumeric registration information from the localized plate crop. EasyOCR is suitable for this stage because it provides a Reader interface and a readtext() method for extracting text from image inputs [28]. In the proposed subsystem, this OCR process transforms the violation from a visual event into an identifiable enforcement record.

However, in the compliant helmet-wearing case, this stage is skipped. This behavior is academically important because it demonstrates decision-based processing. The system does not simply perform detection and recognition sequentially on every input; instead, it uses the helmet violation decision as a control condition for subsequent vehicle identification.

Figure 6.4: Conditional number plate recognition process



6.1.5 Summary of Observed Output Behavior

The observed outputs confirm that the proposed subsystem produces different results depending on helmet compliance status. In the violation case, the system identifies non-compliance and reads the vehicle number plate. In the compliant case, the system correctly suppresses the number plate recognition process. In the webcam case, the subsystem demonstrates its ability to process live input and display a violation result.

Table 6.1: Summary of observed system behavior

Scenario	Helmet Status	Violation Decision	Number Plate Recognition	Expected System Behavior
Case 1	Helmet not worn	Violation detected	Performed	Detect violation and read plate number
Case 2	Helmet worn	No violation	Not performed	Confirm compliance and stop further enforcement processing
Case 3	Helmet not worn in webcam footage	Violation detected	Optional, depending on frame quality and plate visibility	Display violation from live input

6.1.6 Model Evaluation Interpretation

The helmet detection model achieved an overall precision of 0.7207, recall of 0.6737, mAP50 of 0.6879, and mAP50–95 of 0.3284. These results indicate that the model achieved usable detection capability, although its performance requires further improvement for high-stakes enforcement deployment. Precision and recall are important because they indicate the reliability of positive detections and the model's ability to identify relevant objects, while mAP is a widely used object detection metric that evaluates detection quality across classes and intersection-over-union thresholds [27].

The difference between mAP50 and mAP50–95 suggests that the detector performs better under a standard IoU threshold than under stricter localization requirements. This means that the model can often identify the relevant object category, but the bounding-box localization may not always remain accurate under stricter evaluation conditions. In the context of helmet violation detection, this is especially important because the helmet or no-helmet region is relatively small compared with the entire rider or motorcycle.

The number plate detection stage showed stronger performance, with approximately 97.8% precision, 95.7% recall, 96.7% accuracy, and 96.8% F1-score. This indicates that plate localization was more reliable than helmet-status detection. The stronger performance can be attributed to the more consistent rectangular structure of vehicle number plates and the single-class nature of the plate detection task. The component documentation similarly identifies number plate detection as comparatively stronger than no-helmet detection.

Table 6.2: Performance interpretation of the main subsystem modules

Module	Observed Result	Interpretation	Practical Implication
Helmet detection precision	0.7207	Moderate reliability of positive helmet-status predictions	False positives should be reduced before high-stakes use
Helmet detection recall	0.6737	Some helmet-status cases may be missed	Additional training data and class balancing are required
Helmet detection mAP50	0.6879	Usable detection at standard IoU threshold	Suitable for prototype-level validation
Helmet detection mAP50–95	0.3284	Weaker localization under stricter IoU thresholds	Bounding-box accuracy should be improved
Plate detection precision	97.8%	Strong reliability of plate detections	Suitable for violation-linked plate localization
Plate detection recall	95.7%	Most visible plates were detected	Supports vehicle identification

Plate detection F1-score	96.8%	Balanced plate detection performance	Plate detection is a strong subsystem component

6.2 Research Findings

The first major finding is that the proposed system successfully implements conditional enforcement logic. A helmet-wearing rider is classified as compliant, and the number plate recognition stage is not executed. Conversely, when a rider is detected without a helmet, the system classifies the event as a violation and activates number plate recognition. This confirms that the system does not function as a generic number plate reader but as a violation-triggered enforcement-support subsystem.

The second finding is that webcam-based input can be used for helmet violation detection. The webcam result demonstrates that the trained model and decision logic can operate on live camera frames, which is important for practical surveillance deployment. This supports the feasibility of extending the subsystem from offline testing to real-world traffic monitoring scenarios.

The third finding is that number plate recognition becomes operationally meaningful only when linked to a violation event. In isolation, number plate recognition only identifies a vehicle. In the proposed system, however, number plate reading is connected to a helmet violation decision, thereby producing an actionable enforcement output. This linkage is one of the main contributions of the component.

The fourth finding is that the no-helmet class remains more challenging than plate localization. This is likely due to the small visual size of the rider's head region, occlusion, motion blur, camera angle, and possible class imbalance in the dataset. The component documentation also identifies no-helmet detection as the weakest class and highlights sensitivity to video quality, camera position, and occlusion.

6.3 Discussion

The results demonstrate that the proposed subsystem can support automated helmet violation monitoring by integrating helmet-status detection, violation decision-making, conditional plate recognition, and live camera-based output. This integration is important because automated enforcement requires more than object detection. A practical system must determine whether a violation has occurred, avoid unnecessary processing for compliant cases, and associate confirmed violations with vehicle identity.

The non-compliant case demonstrates the complete enforcement path of the subsystem. When the rider is not wearing a helmet, the system classifies the case as a violation and proceeds to number plate recognition. This confirms that the system can generate an enforcement-oriented output by linking rider safety non-compliance with vehicle registration information. The result is consistent with the stated purpose of the component: to transform raw traffic surveillance footage into actionable enforcement evidence.

The compliant case is equally important because it validates the negative decision path. A system that reads every number plate regardless of violation status would be inefficient and would generate irrelevant enforcement records. By stopping the pipeline when helmet compliance is detected, the subsystem demonstrates controlled and purpose-specific processing. This behavior improves the practical suitability of the system for deployment in traffic monitoring environments.

The webcam-based output demonstrates that the subsystem can be extended beyond offline image or video testing. Real-time camera input introduces additional complexity because frames may vary in lighting, angle, motion, and clarity. The successful display of a no-helmet violation from webcam footage therefore provides initial evidence that the system can function under live-input conditions. However, broader validation using real roadside surveillance cameras would be required before deployment.

The major limitation remains the detection of the no-helmet class. Unlike number plates, which usually have a consistent rectangular shape, helmet-status recognition depends on small and variable visual features around the rider’s head. The model may therefore struggle when the head region is blurred, partially occluded, or captured from an unfavorable angle. Future improvements should include additional no-helmet samples, diverse lighting conditions, multiple camera perspectives, improved annotation consistency, and duplicate-frame suppression for video streams.

Overall, the results show that the proposed component contributes meaningfully to an intelligent traffic violation management framework. Its strongest contribution is the conditional integration of helmet violation detection and number plate identification. The system does not merely identify objects; it applies decision logic to determine whether enforcement processing should continue. This makes the component suitable as an applied computer vision subsystem for automated, evidence-based traffic law enforcement.

7. Conclusion

This research presented an intelligent computer vision-based subsystem for automated helmet violation detection and vehicle number plate identification within an integrated traffic violation management framework. The study was motivated by the need to improve the efficiency, consistency, and traceability of traffic law enforcement, particularly in relation to motorcycle helmet compliance. Conventional traffic monitoring practices often depend on manual observation, delayed reporting, and fragmented evidence handling, which can limit the timely identification of violations and reduce the effectiveness of enforcement procedures. In response to these limitations, the proposed subsystem was designed to process traffic surveillance input, determine helmet compliance, identify non-compliant riders as violation cases, and associate confirmed violations with vehicle registration information through number plate detection and recognition.

The main objective of this research component was to develop a practical and automated approach for detecting motorcycle riders who are not wearing helmets and linking such violations with identifiable vehicle information. This objective was achieved through the design and implementation of a sequential detection pipeline consisting of helmet status detection, violation decision-making, evidence capture, number plate localization, and plate text extraction. The system was developed using deep learning-based object detection techniques, with YOLOv8 employed for both helmet-related detection and number plate localization. Optical character recognition was incorporated to extract alphanumeric plate information from detected plate regions. The integration of these stages enabled the subsystem to move beyond isolated object detection and function as an enforcement-support mechanism capable of producing actionable outputs.

The experimental results demonstrated that the proposed system is capable of identifying helmet compliance and non-compliance from traffic surveillance input. In cases where a rider was detected without a helmet, the system classified the event as a violation and proceeded to detect and read the vehicle number plate. Conversely, when the rider was detected as wearing a helmet, the system correctly treated the case as non-violating and did not activate the number plate recognition stage. This conditional processing logic is one of the central contributions of the research, as it ensures that vehicle identification is performed only when an enforceable violation has been detected. Such a design reduces unnecessary processing, avoids irrelevant plate reading for compliant riders, and improves the operational relevance of the subsystem.

The webcam-based result further demonstrated the feasibility of applying the proposed approach to live or camera-based input. This is significant because practical traffic enforcement environments frequently depend on real-time or near-real-time surveillance streams rather than static image analysis alone. The ability of the subsystem to process webcam footage and identify helmet violations indicates that the proposed method can be extended toward real-time monitoring scenarios. Although further validation under diverse roadside conditions is required, the webcam-based output provides initial evidence that the system is suitable for practical deployment-oriented development.

The number plate identification stage also contributed significantly to the overall value of the proposed subsystem. A helmet violation detection model, when used independently, can identify unsafe rider behavior but cannot support enforcement unless the detected violation is linked to a specific vehicle. By incorporating number plate detection and recognition, the system provides a connection between the violation event and vehicle identity. This makes the output more useful for downstream enforcement procedures, database storage, evidence management, and review by relevant authorities. Therefore, the integration of helmet violation detection with number plate identification strengthens the practical applicability of the research within intelligent transportation systems.

The evaluation results indicated that the number plate detection component performed more strongly than the helmet violation detection component. This can be attributed to the relatively structured and consistent visual appearance of number plates, as well as the single-class nature of the plate localization task. In contrast, helmet-status detection, particularly the identification of riders without helmets, was more challenging due to small visual regions, partial occlusion, rider posture, motion blur, camera angle, and variations in lighting. This finding is important because it highlights that the most enforcement-critical class is also one of the most difficult to detect reliably. The research therefore demonstrates not only the feasibility of the proposed approach but also the technical challenges that must be addressed before large-scale deployment.

A key strength of this research is the development of a complete violation-oriented workflow. The subsystem does not merely generate class labels or bounding boxes; it applies decision logic, captures evidence, activates number plate recognition conditionally, and supports the generation of structured violation outputs. This makes the proposed component suitable for integration into a broader smart traffic violation management system. The research also contributes to the wider objective of automated, data-driven traffic governance by supporting faster violation identification, improved evidence handling, and reduced dependency on continuous manual monitoring.

Despite these contributions, several limitations were identified. The detection of the no-helmet class requires further improvement to ensure greater robustness under complex real-world conditions. Surveillance footage may include low-resolution frames, crowded traffic scenes, poor illumination, side-view perspectives, and occlusions caused by vehicles or other riders. These factors can reduce detection accuracy and may lead to missed detections or incorrect classifications. In addition, repeated detections across consecutive video frames may generate duplicate evidence records if temporal filtering or tracking is not applied. The OCR stage is also sensitive to plate quality, including blur, reflection, viewing angle, dirt, and non-standard plate visibility. These limitations indicate the need for further refinement before the subsystem can be deployed in high-stakes enforcement environments.

Future work should focus on expanding the dataset with more diverse real-world traffic images and videos, particularly including difficult no-helmet cases captured under different lighting, weather, road, and camera-angle conditions. The helmet detection model could be improved through class balancing, improved annotation consistency, additional training epochs, and exploration of alternative or larger detection architectures. For video-based operation, object tracking and duplicate-suppression mechanisms should be incorporated to prevent repeated violation records for the same rider. The number plate recognition stage can be strengthened through image enhancement, character-level validation, repeated-frame voting, and format-based verification for local number plate structures. Furthermore, integration with a centralized database, authority review dashboard, and citizen appeal mechanism would improve the transparency and completeness of the overall traffic violation management framework.

In conclusion, this research successfully developed and evaluated a helmet violation detection and vehicle number plate identification subsystem that transforms traffic surveillance input into enforcement-oriented evidence. The system demonstrated the ability to distinguish between helmet-wearing and non-helmet-wearing riders, activate number plate recognition only in violation cases, and support vehicle identification for non-compliant events. While further improvements are required to enhance robustness under real-world conditions, the proposed subsystem provides a strong foundation for automated traffic law enforcement. It contributes to the broader development of intelligent transportation systems by combining deep learning-based detection, conditional decision-making, evidence generation, and vehicle identification within a coherent and practically meaningful framework.

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